

Physiologically-based Pharmacokinetic – Informed New Approach Methodology Framework Modeling to Guide Reduced Monkey PK Studies for Monoclonal Antibodies: A Tutorial

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Conflict of Interest

Stephan Schaller is founder and employee of ESQlabs, the company that developed the R package esqlabsr used in this tutorial.

Abstract

Objectives: This tutorial presents a reproducible PK-Sim/MoBi-R workflow for predicting monkey and human monoclonal antibody (mAb) pharmacokinetics (PK) is governed by mechanistic processes and physiologically-based pharmacokinetic (PBPK) models are well suited for mAb's human PK prediction. We evaluated whether without calibration to in vitro-informed "a priori" PBPK modeling could predict first-in-human mAb PK-vivo PK profiles and support for evaluating reduced monkey PK-study designs as within a new approach methodology (NAM). A semi-automated PK-Sim/MoBi-R workflow integrated species framework.

Methods: Species physiology with, FcRn-binding, and target-mediated disposition inputs. Intravenous were integrated to predict intravenous single-dose PK data for eight mAbs across five targets were evaluated retrospectively. "A priori" predictions, generated without fitting to monkey PK data, Predicted AUC_{0-tlast}, C_{max}, and clearance were compared with observed AUC_{0-28d}, C_{max}, and clearance and with predictions from monkey-fitted top-down models translated to humans. Reduced values. Candidate monkey designs for golimumab and nivolumab were evaluated under 30% and 60% CV scenarios, incorporating uncertainty in five mechanistic inputs and between-subject variability in body weight and target concentration. Sparse-study designs were evaluated using simulations means were compared primarily with replicate-matched dense-population means.

For human PK, 83.3%, 96.7%, and 76.7% of "a priori" AUC_{0-28d}, C_{max}, and clearance predictions, respectively, were within two-fold of observed values, compared with 86.7%, 90.0%, and 93.3% for top-down translation. A two dose groups design with three animals per group (total n=6) suggested a success probability of 99.4-100% under 30% parameter variability and estimated uncertainty. The predicted success probability declined to 90.0-98.1% with 50% variability assumption.

These findings support further evaluation of "a priori" PBPK models informed NAM workflow for first-in-human mAb PK prediction. Predictive performance of the "a priori" approach was broadly comparable to monkey-to-human translation, although clearance predictions were sensitive to uncertainties in parameters and in available data. Under the stated assumptions, the reduced design required six rather than at least 15 animals, a 60% reduction at the individual PK-study level and suggests prospective validation.

Results: AUC_{0-tlast} and C_{max} were predicted within two-fold of observed values at all evaluated doses for five of eight mAbs. Across individual profiles, the proportions within two-fold for AUC_{0-tlast}, C_{max}, and evaluable clearance were 85.7%, 92.9%, and 50.0% in

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monkeys and 96.7%, 96.7%, and 66.7% in humans, respectively. Study-design simulations indicated that two dose groups of two monkeys each (n=4) represented the minimum adequate design. Increasing the group size to three monkeys (n=6) modestly improved precision, whereas adding further dose groups or animals per group yielded only marginal gains.

Conclusions: This tutorial provides a reproducible workflow for predicting IV mAb PK for monkey and human without calibrating with in vivo data and testing smaller exploratory monkey PK study designs, thereby offering a framework for prospective evaluations using larger datasets and experimentally measured mechanistic inputs.

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Introduction

Adequate characterization of pharmacokinetics (PK) is crucial for a successful drug development program because it is the quantitative link between administered dose, systemic and tissue exposure, target engagement, pharmacodynamic (PD) response, and safety [1]. For monoclonal antibodies (mAbs) and other protein therapeutics, systemic and tissue exposures depend on target-mediated drug disposition (TMDD), neonatal Fc receptor (FcRn)-mediated recycling, and tissue distribution processes that can differ across species. Physiologically-based pharmacokinetic (PBPK) modeling is a mechanistic modeling approach that integrates species-specific physiology with drug- and target-specific parameters to simulate PK. For mAbs, PBPK models incorporate key determinants of disposition, including vascular convection, lymphatic return, tissue partitioning, pinocytosis, FcRn binding and recycling, proteolytic catabolism, target binding, target turnover, and target-receptor complex internalization, into a quantitative framework [2,3,4]. Overall, extensive work on PBPK models for mAbs has already established a robust scientific foundation, enabled reliable characterization of their PK, and supported their regulatory acceptance.

Momentum for new approach methodologies (NAMs) has increased exponentially since the FDA's April 2025 Roadmap to Reducing Animal Testing in Preclinical Safety Studies [5, 6, 7, 8, 9]. Within this evolving regulatory infrastructure, current NAM-focused investment has focused on establishing context-of-use (COU) frameworks, advancing complex in vitro systems (organoids, microphysiological systems, organ-on-chip platforms), and AI/ML-based computational toxicology tools. On the other hand, preclinical evaluations are increasingly being framed around whether human-relevant biological pathways are perturbed at clinically meaningful plasma and tissue exposure levels. Plasma and tissue exposures therefore provide a common quantitative measure for integrating in vitro dose-response data, in silico simulations, prior clinical knowledge, biomarkers, and safety thresholds. For example, in vitro findings from cell-based or microphysiological systems can be contextualized through predicted human concentration-time profiles, exposure margins, and time above biologically active or adverse effect thresholds [10, 11]. In this context, mechanistic PBPK models are uniquely positioned to bridge NAM-derived biological observations with quantitative predictions of human plasma and tissue exposure. A recent FDA/CDER landscape analysis of 15 years of NAM-related submissions found that 93% fell into the in vitro and in silico categories broadly, though this landscape analysis does not break out mechanistic PK/PD/safety modeling, such as PBPK, as distinct categories, leaving their specific representation within the NAM submission landscape unclear [12]. By contrast, mechanistic modeling approaches, such as, PBPK, quantitative systems pharmacology (QSP), and quantitative systems toxicology (QST), despite an

established and steadily growing regulatory submission record under the parallel model-informed drug development (MIDD) paradigm and demonstrated translational utility [8, 13], have received comparatively limited dedicated attention within the emerging NAM ecosystem, representing both a gap and an opportunity. Overall, the emerging NAM ecosystem presents a significant opportunity for greater integration of mechanistic modeling approaches.

Focusing on mAb PK characterization and the role of mechanistic PK modeling, although a scientific framework for mAb PBPK exists, preclinical PK characterization mostly relies on PK studies in monkeys (often cynomolgus monkeys). A traditional mAb development program typically includes preclinical PK evaluations in at least two studies: an exploratory single-dose PK study (n = 12-18 monkeys) and a definitive repeat-dose PK study (n = 12-24 monkeys). Together, these studies generally require a minimum of 24-42 monkeys per mAb development program. The scientific maturity of mAb PBPK modeling has advanced to the point where “a priori” PK predictions based on in vitro data can be considered for prospective evaluation in preclinical development workflows, as NAMs. Published examples have shown that mAb-specific in vitro measurements, together with species physiology and FcRn-mAb interaction-related parameters, can support prediction of mAb linear PK without in vivo data [14, 155-6]; however, these frameworks did not incorporate target-mAb interactions governing nonlinear, target-mediated clearance. Although some mAbs exhibit linear PK at dose range of interest, target-mediated drug disposition is inherent to therapeutic antibodies. For example, at high doses, even mAbs known to exhibit non-linear PK can appear linear upon target saturation, and at low doses, even mAbs known as linear may show nonlinearity. Overall, extending “a priori” PBPK frameworks to incorporate target-mediated disposition is therefore important for their broader application as NAMs with the potential to reduce preclinical monkey studies across the full range of mAb PK profiles. This foundation provides an opportunity to incorporate established PBPK and TMDD methods within a new approach methodologies (NAMs) decision framework.

While complete replacement Interest in NAMs has increased substantially following the FDA's April 2025 Roadmap to Reducing Animal Testing in Preclinical Safety Studies [7-11]. Current NAM efforts encompass context-of-use (COU) frameworks, complex in vitro systems, including organoids, microphysiological systems, and organ-on-chip platforms, and AI/ML-enabled computational toxicology. These approaches increasingly evaluate whether human-relevant biological pathways are perturbed at clinically meaningful plasma or tissue exposures. Exposure therefore provides a common quantitative basis for integrating in vitro concentration-response data, in silico predictions, prior clinical knowledge, biomarkers, and safety thresholds. For example, findings from cell-based or microphysiological systems can be interpreted in the context of predicted human

concentration-time profiles, exposure margins, and the duration for which concentrations exceed biologically active or adverse-effect thresholds [12-13].

Despite the growing interest, the role of quantitative pharmacology within the NAM landscape remains insufficiently delineated. A recent FDA/CDER analysis of 15 years of NAM-related submissions reported that 93% involved broadly defined in vitro or in silico approaches, but did not clearly identify examples of mechanistic quantitative pharmacology approaches, such as physiologically based pharmacokinetic (PBPK), quantitative systems pharmacology (QSP), or quantitative systems toxicology (QST) modeling [14]. Consequently, the extent to which these model-informed drug development (MIDD) approaches are being applied specifically as NAMs remains unclear, despite their established and growing use in regulatory submissions [10, 15]. Mechanistic quantitative pharmacology can play an important role at the interface between preclinical evidence and clinical translation by connecting NAM-derived biological observations to predictions of clinically relevant plasma and tissue exposures.

Here, we illustrate this role through a practical workflow that applies in vitro data, PBPK modeling, TMDD principles, and study simulations to inform the design of mAb PK studies. ~~The example focuses on optimizing monkey PK studies-is not yet scientifically justifiable; model-informed optimization of study design offers a pragmatic reduction strategy. Prior knowledge of the mechanistic determinants of mAb disposition, PBPK predictions, and in vitro data can inform the selection of dose, including dose groups, animals, and sampling times-needed animal numbers, to address specific PK uncertainties, complementing approaches defined uncertainty while reducing animal use. The same workflow can be adapted to other types of preclinical studies and can complement strategies such as cassette dosing, microsampling, animal reuse, and integration of PK/PD or toxicology endpoints into existing studies [16-18]. Because regulatory expectations and molecule-specific uncertainty vary, simulations should inform rather than independently define the minimum acceptable study size. It can also support evaluation of alternatives to nonhuman primates, including hybrid approaches that scale human PK from transgenic human FcRn mice using allometric or PBPK methods. This article presents a reproducible, tutorial-oriented workflow for applying established PBPK and TMDD methods within a NAM decision framework for mAb development. The primary objectives are (1) to demonstrate how to generate monkey and human PBPK predictions in PK-Sim/MoBi without calibration to the analyzed in vivo PK profiles and (2) to explore opportunities to reduce animal numbers in monkey PK studies using those models. A secondary objective is to evaluate the workflow retrospectively using eight intravenously administered mAbs across five targets. The analysis assesses predictive performance and uses simulations to show how uncertainty in mechanistic inputs and interindividual variability affects confidence that~~

candidate designs can recover model-defined PK characteristics. These examples are intended to help preclinical pharmacology and modeling teams develop models, interpret outputs, and design fit-for-purpose PK studies with fewer dose groups and animals while identifying the additional evidence needed for prospective application.

Overview

Figure 1 summarizes the four-part workflow used to develop and evaluate the PBPK models. First, six drug- and target-specific parameters were obtained from in vitro studies or the literature to provide independent mechanistic inputs: antibody–FcRn binding affinity, antibody–target binding affinity and dissociation rate, target–antibody complex internalization rate, target degradation rate, and target reference concentration. Published intravenous plasma PK profiles for eight mAbs targeting five proteins in monkeys and humans were digitized using WebPlotDigitizer. Most reports did not specify whether concentrations represented free or total antibody. Literature sources also provided the drug–FcRn, drug–target, complex internalization, target degradation, and target concentration inputs. Target concentration and degradation values were shared among mAbs directed against the same target. Inputs derived from previous modeling studies rather than independent in vitro measurements are identified in Supplementary Material 2 and were considered when interpreting predictive performance.

Second, species- and target-specific models were constructed in PK-Sim and extended in MoBi with target-mediated drug disposition reactions describing complex internalization and target degradation. The resulting models were exported as compiled PKML files so that R-generated configuration files could apply the appropriate molecule-, species-, dose-, and administration-specific inputs consistently. Third, these configurations were used to predict monkey and human PK without calibration to observed data. Predicted concentration–time profiles and PK endpoints (C_{max} , $AUC_{0-t_{last}}$, AUC_{inf} , and clearance) were then compared retrospectively with observations using predicted-to-observed fold errors and a prespecified 0.5- to 2-fold criterion. The fully reproducible workflow and instructions are provided in Supplementary Material 1.

The fourth part evaluated whether smaller exploratory monkey studies could reliably recover “true PK” endpoints under defined assumptions about parameter uncertainty and population variability. Reference models for golimumab and nivolumab generated densely sampled profiles for 50 monkeys across five doses. For each molecule and variability scenario (30% or 60% CV), 100 Monte Carlo replicates were performed. Within each replicate, one lognormal draw for each of five uncertain model parameters was shared across all monkeys and doses, whereas body weight and target reference concentration

varied among monkeys. This generated 100,000 population profiles for evaluating candidate designs with two, three, or five dose groups; two, three, or six monkeys per group; and sparse sampling through day 28. For the primary analysis, each sparse study-group mean was compared with the mean of the corresponding 50 replicate-matched dense individual endpoints; a secondary analysis instead used fixed nominal typical-subject endpoints. Fold-error distributions for AUC_{0–28d}, C_{max}, and clearance were evaluated against the prespecified 0.5- to 2-fold criterion.

Software: PK-Sim/MoBi and the R packages `esqlabsR` and `ospsuite` provide a practical open-source framework for a reproducible, semi-automated workflow [19–21]. The platform combines physiologically structured PBPK models with curated species-specific physiology and expression databases, allowing target-expression profiles to be defined consistently across monkeys and humans [22–23]. Export to MoBi and programmatic simulation in R also enable scalable evaluation across molecules, targets, species, and dose levels. This reproducibility is important for prospective PBPK-based NAM workflows, in which model generation, parameter updates, simulations, and exposure-metric extraction should be prespecified and independently traceable.The objectives of the current analysis were (1) to retrospectively evaluate “a priori” PBPK modeling as a NAM for predicting human PK of intravenously administered mAbs and (2) to conduct a model-conditional precision analysis of reduced exploratory single-dose monkey PK study designs with fewer dose groups and animals per group.

Methods

Data

Pharmacokinetic profiles for eight monoclonal antibodies, spanning across 5 different targets, following intravenous dosing in monkeys and in humans were digitized from literature using `webplotdigitizer` tool [<https://automeris.io/>]. For most of the monoclonal antibodies included in the analysis, the literature sources did not specify whether the digitized concentration data represented free or total antibody. In vitro experimental parameters relevant to drug-FcRn binding, drug-target binding, drug-target complex internalization rate, and target reference concentrations were obtained through literature search. Digitized data, parameter values, measurement methods, assumptions, and source references are provided in Supplementary material 2.

“A priori” The example workflow was implemented with PK-Sim 12.2 and MoBi 12.2 for model construction, R 4.5.1 for programmatic execution and analysis, `esqlabsR` 5.6.0 and

osp suite 12.4.2 for the R interface to the Open Systems Pharmacology Suite, and Quarto for the executable tutorial documents [19-21, 24].

Generating PBPK Predictions Without In Vivo PK Calibration

Configuring Target-Specific PBPK Approach Models

In this tutorial, we began with the generic large-molecule structural model ~~was first incorporated~~ in PK-Sim, including its binding to FcRn and to a generic target [4]. The model was exported to MoBi and TMDD processes were added using a modular approach [19, 25]. Next, we created target expression profiles in monkeys and humans for each of the targets, including relevant localization, using PK-Sim's expression database [20, 22-23]. The expression database provides relative abundance of target expression across various tissues. Absolute reference concentrations for each of the targets, i.e., concentrations of the target, that is, the concentration in the tissue with the highest expression, were set to literature-derived values and are provided in S1.3. Next, PKML simulation files containing generic large-molecule PK model and target expressions, one file per target and species (monkey or human), were exported [19]. Intravenous single-dose simulations were run programmatically, and C_{max}, AUC_{0-28d}, clearance, and fold errors were calculated for each molecule-dose pair with observed data. C_{max} and AUC_{0-28d} were used as exposure metrics. Clearance was calculated as dose divided by AUC_{inf} and represents total systemic clearance; at target-saturating doses it is expected to be dominated by linear clearance rather than serving as a specific marker of nonlinear target-mediated elimination. The percentage of AUC_{inf} extrapolated was calculated as $100 \times (C_{last}/\lambda_z)/AUC_{inf}$, equivalent to $100 \times (AUC_{inf} - AUC_{last})/AUC_{inf}$, and values greater than 20% were flagged in the separate analysis tables. Because AUC_{0-28d} contributes to AUC_{inf} and clearance is calculated from AUC_{inf}, AUC_{0-28d} and clearance are mathematically related and were not interpreted as independent endpoints. The workflow used PK-Sim 12.2, MoBi 12.2, R 4.5.1, esqlabsR 5.6.0, and OSP Suite 12.4.2 [22-24]. Full workflow details, input files, and code are provided in Supplementary Material 2-derived values and are provided in Supplementary Material 2. PKML simulation files containing the generic large-molecule PK model and target expression, one file per target and species, were then exported [25]. The 01_Models folder in Supplementary Material 1 documents the model-building steps and supplies the files used in the executable workflow.

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Top-Down TMDD Approach

In the top-down approach, a quasi-steady-state (QSS) TMDD model was fitted separately to the monkey PK profiles for each molecule in R 4.5.1 using nlmixr2 5.0.0 and rxode2 5.1.4. Clearance (CL), central volume (V1), intercompartmental clearance (Q), peripheral volume (V2), and proportional and additive residual error terms were estimated when supported by the data. Molecule-specific TMDD parameters, including target reference concentration, binding kinetics, target degradation rate (kdeg), and internalization rate constant (kint), were fixed to literature-derived monkey inputs. The same molecule-specific inputs were used in the “a priori” monkey PBPK simulations and the top-down monkey fits.

Human PK predictions were generated by translating the final Run 4 monkey TMDD model to humans. Monkey CL and Q estimates were allometrically scaled with an exponent of 0.85, while Vc and Vp were scaled with an exponent of 1.0. All molecule-specific TMDD inputs used in the human simulations, including target reference concentration, binding affinity (KD), dissociation rate (koff), target degradation rate (kdeg), and internalization rate constant (kint)—were obtained from the human parameter dataset, and the human QSS binding constant was recalculated from the human kinetic inputs. Simulated monkey and human Cmax, AUC_{0-28d}, and clearance values were compared with observed values, and fold-errors were calculated.

Running Simulations-based Evaluations of Reduced Monkey PK Study Designs

For molecules with at least one “a priori” monkey exposure prediction outside the two-fold criterion, reduced study simulations were performed as an explicitly model-conditional precision analysis. Because individual-level observed monkey PK data were unavailable, the observed data-informed top-down model served as both the data-generating model and the retrospective reference. For each molecule, 100 virtual individuals were generated using the final Run 4 CL, V1, Q, and V2 estimates and their RSEs; 40% parameter uncertainty was assigned when a fixed structural parameter had no fitted RSE (S1.4). Independent lognormal 30% interindividual variability was applied to CL, V1, Q, and V2, while the fixed TMDD parameters were unchanged. One thousand replicate studies sampled 2-4 animals per dose group with replacement at 0.25, 1, 2, 4, and 8 hours and days 1, 2, 6, 10, 14, 21, and 28. Two-dose (0.3 and 10 mg/kg) and three-dose (0.3, 3, and 10 mg/kg) designs were evaluated. The arithmetic mean Cmax, AUC_{0-28d}, and clearance in each reduced study was divided by the corresponding arithmetic mean from the richly sampled 100-animal population. A design-level endpoint was successful only when all selected doses were within 0.5- to 2-fold of the reference. In a post hoc sensitivity analysis, additional independent lognormal variability was applied to CL, V1, Q, and V2 so that the

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total across-individual CV of each structural parameter was at least 50%; 10,000 reduced studies were resampled per scenario.

Results

Evaluation of “A Priori” PBPK Predictions of Monkey and Human PK

First, both “a priori” PBPK and top-down TMDD approaches were compared to evaluate how well each workflow reproduced the monkey PK data. Across the 8 drugs and 14 dose levels evaluated, the “a priori” PBPK predictions produced 8/14 AUC_{0-28d} (accuracy 0.57), 13/14 C_{max} (accuracy 0.93), and 8/14 clearance (accuracy 0.57) within two-fold of the observed (**S1.1A**, **S1.2**). As a benchmark comparison, with top-down approach, as expected, fitting a quasi-steady-state (QSS) TMDD model to monkey PK data while fixing molecule-specific TMDD parameters produced all AUC_{0-28d} , C_{max} , and clearance values within two-fold of the observed data (accuracy 1.00 for each metric). CL and V_c were estimated for all molecules. Intercompartmental clearance was fixed to a typical cynomolgus-monkey mAb value for five molecules for which it could not be estimated with acceptable precision; peripheral volume was additionally fixed for adalimumab [25]. All remaining estimated structural parameters had RSEs <40%, and the covariance step completed successfully for all final Run 4 fits (**S1.4**).

Next, both “a priori” PBPK and monkey-informed top-down TMDD approaches were compared to evaluate how well each workflow predicted human PK data. Across the 8 drugs and 30 dose levels evaluated, the “a priori” PBPK predictions produced 25/30 AUC_{0-28d} (accuracy 0.83), 29/30 C_{max} (accuracy 0.97), and 23/30 clearance values (accuracy 0.77) within two-fold of the observed data (**Figure 1**, **Figure 2**). With the top-down approach, translation of the monkey-fitted QSS TMDD models to humans produced 26/30 AUC_{0-28d} (accuracy 0.87), 27/30 C_{max} (accuracy 0.90), and 28/30 clearance (accuracy 0.93) within two-fold (**Figure 1**, **Figure 2**).

The “a priori” PBPK framework predicted AUC_{0-28d} and C_{max} for 3/8 molecules, and all three exposure markers for 4/8 molecules across all doses evaluated. Compared with the top-down approach, the “a priori” approach provided comparable accuracy for AUC_{0-28d} and C_{max} , but lower accuracy for clearance (0.77 versus 0.93). The molecules for which human clearance was predicted outside the two-fold criterion by the “a priori” approach were avetumab (anti-PD-L1 mAb), pembrolizumab (anti-PD-1 mAb), and gotimumab (anti-TNF α mAb). For these molecules, all C_{max} predictions were within two-fold of the observed values across the evaluated doses. AUC_{0-28d} predictions were also within two-fold for both pembrolizumab dose levels and five of the six gotimumab dose levels, but for only

one of the four avelumab dose levels. Thus, the prediction discrepancies were concentrated in clearance-related metrics rather than G_{max} , with the greatest AUC_{0-28d} discrepancy observed for avelumab. For some molecules in the dataset, several key target-system parameters, including target reference concentration, internalization rate (k_{int}), and target degradation rate (k_{deg}), were not available from direct in vitro measurements and were instead obtained from prior modeling studies. These inputs introduce uncertainty (S1.6) because their values may depend on the model structure, assumptions, and estimation approach used in those studies. Additionally, the observed concentration-time data through 28 days post-dose may not have adequately characterized the terminal phase, limiting the reliability of the observed clearance estimates, as the extrapolated fraction of AUC_{inf} exceeded 30% for several molecules (S1.5). Moreover, the observed human PK data for oncology mAbs, including avelumab, pembrolizumab, and nivolumab, were obtained from patients who may have substantial variability in target expression, introducing further uncertainty in PK predictions with both approaches. Overall, the “a priori” approach predicted mechanisms of mAbs distribution and elimination in humans reasonably well.

Simulations-based Evaluations of Reduced Monkey PK Study Designs

Atezolizumab, avelumab, nivolumab, and pembrolizumab were included in the reduced-study analysis because at least one “a priori” monkey exposure prediction was outside the prespecified two-fold criterion. For each molecule, the observed data-informed top-down model generated a richly sampled reference population of 100 virtual individuals. The primary 1,000-replicate analysis incorporated fitted structural-parameter uncertainty and 30% interindividual variability in CL, V1, Q, and V2; fixed TMDD parameters were not varied.

Across the two-dose-group (0.3 and 10 mg/kg) and three-dose-group (0.3, 3, and 10 mg/kg) designs with 2–4 animals per group, reduced-study-to-reference fold errors for AUC_{0-28d} , G_{max} , and clearance were centered near 1 (Figure 3). For the illustrative two-dose-group design with three animals per group (six animals total), the model conditional probability that all three PK endpoints at both doses were within two-fold of the rich-population reference ranged from 99.4% to 100.0% across the four molecules. The 5th–95th percentile fold-error intervals also remained within the prespecified range. A fourth animal per group provided only modest additional narrowing, and adding a third dose group did not improve precision under the evaluated assumptions. The results were sensitive to greater variability: for the same six-animal design, the joint success probability decreased to 90.0–98.1% when total structural-parameter variability was increased to 50% CV. Overall, these findings support the feasibility of a reduced two-dose, total six-animal study design for

collecting relevant data to support model-based characterization of IV mAbs PK in monkeys to support first-in-human PK predictions.

Discussion

In this retrospective analysis, the in vitro-informed “a priori” PBPK approach provided human PK predictions broadly comparable to monkey-informed top-down TMDD translation, particularly for AUC_{0-24h} and G_{max} . Clearance accuracy was lower with the “a priori” approach, with 23/30 predictions within two-fold of observed values compared with 28/30 for the top-down approach. Notably, target concentrations, k_{tr} , and k_{deg} for pembrolizumab, nivolumab, and avelumab were obtained from prior models rather than molecule-specific in vitro measurements, suggesting that robust in vitro characterization of these parameters could further improve “a priori” predictions of clearance (S1.6). The proposed “a priori” PBPK modeling as NAM approach is feasible when robust in vitro experimental data for mechanistic parameters are available. In our retrospective analysis, in vitro parameters were derived from literature search (Supplementary material 2). Drug-target binding affinity was available from experimental binding assays for all included molecules, either from equilibrium binding or SPR/Biacore assays. FcRn binding affinity was available from SPR/Biacore for only atezolizumab and pembrolizumab while a common model-based FcRn KD estimate from a published analysis of 50 mAbs was used for several other mAbs where molecule-specific FcRn binding data were not available [27]. Drug-target complex internalization rate from flow-cytometry-based internalization assay was available for only two molecules, whereas for other molecules this parameter was assumed same as target degradation rate. Target reference concentrations represented the least standardized component of the input parameters. VEGF-A and IL-6R reference concentrations and target degradation rates were available from quantitative fluorescence or ligand-binding receptor assay data or were estimated in prior analyses. Overall, the compiled input parameters highlight the limitation of standardized in vitro experimental methods. Thus, standardization of experimental package for mAbs would further strengthen prospective PBPK-based NAM workflows.

The reduced-study simulations provide quantitative support for the feasibility of a focused two-dose, six-animal exploratory monkey PK study. Under the primary assumptions of fitted structural-parameter uncertainty and 30% interindividual variability in CL, V_1 , Q , and V_2 , a design with three animals at each of two dose levels recovered G_{max} , AUC_{0-24h} , and clearance within two-fold of the rich-population reference in 99.4–100.0% of replicate studies across the four mAbs. The design remained robust when total structural-parameter variability was increased to 50% CV, with success rates of 90.0–98.1%. These findings should be interpreted as evidence of model-conditional precision rather than a validation

of the reduced design. Prospective confirmation and molecule-specific assessment would therefore be needed, particularly when TMDD is pronounced or its determinants are uncertain. Overall, relative to the prespecified exploratory single-dose design of five dose groups with three animals per group, the illustrative six-animal design represents a 60% reduction in animal use (**Figure 4**). Importantly, this estimate applies at the individual PK-study level and should not be interpreted as a percent reduction in animal use across the overall preclinical development program or as a reduction in animals used for repeat-dose toxicology or other studies.

The intravenous single-dose simulations for each of the eight molecules were run programmatically for monkeys and humans through 2,880 hours, which covered the latest observed sampling time. C_{max} , $AUC_{0-tlast}$, clearance, and fold errors were calculated for each molecule-species-dose profile with observed digitized data. For each observed profile, tlast was the last quantifiable sampling time. The corresponding predicted profile was interpolated to the observed sampling times, and observed and predicted $AUC_{0-tlast}$ were calculated over the identical interval from time zero to tlast using the same linear-up/log-down trapezoidal rule. When no time-zero concentration was available, both profiles were anchored at zero concentration at time zero. AUC_{inf} was calculated as AUC to the last positive concentration plus the last concentration divided by the terminal rate constant (λ_z); λ_z was selected from consecutive post- C_{max} points using the fit with the largest adjusted R-squared and at least three positive concentrations. Clearance was calculated as dose divided by AUC_{inf} and reflects total systemic elimination, including any target-mediated contribution. Observed clearance was considered evaluable only when the extrapolated fraction of AUC_{inf} was no greater than 20%. Profiles above this threshold, or for which the extrapolated fraction could not be estimated, were marked not evaluable and excluded from clearance accuracy calculations. The 02_APriori_PBPK_Workflow folder in **Supplementary Material 1** provides the detailed steps and executable code for the predictions and endpoint calculations.

Predictive Performance

Figure 2 compares PBPK predictions generated without calibration to the analyzed in vivo PK profiles with the corresponding observed monkey and human data. For monkeys, across the eight drugs and 14 dose levels evaluated, 12/14 $AUC_{0-tlast}$ predictions (85.7% accuracy) and 13/14 C_{max} predictions (92.9% accuracy) were within two-fold of the observed values. Observed clearance was evaluable for six monkey scenarios, of which 3/6 predictions (50.0%) were within two-fold. For humans, across the eight drugs and 30 dose levels evaluated, 29/30 $AUC_{0-tlast}$ predictions (96.7% accuracy) and 29/30 C_{max} predictions (96.7% accuracy) were within two-fold. Observed clearance was evaluable for 15 human scenarios, of which 10/15 predictions (66.7%) were within two-fold. The remaining eight

monkey and 15 human clearance scenarios were not evaluable because the observed AUC_{inf} extrapolated fraction exceeded 20% or could not be estimated (Supplementary Material 2).

The PBPK framework predicted both $AUC_{0-tlast}$ and C_{max} within two-fold of the observed values across all evaluated doses for five of the eight molecules. For human PK, $AUC_{0-tlast}$ was within two-fold for every dose except one of the four avelumab doses, and C_{max} was within two-fold for every dose except the highest nivolumab dose. Human clearance was evaluable for 15 of 30 scenarios. Overall, the PBPK approach reasonably predicted observed $AUC_{0-tlast}$ and C_{max} , while the clearance analysis was limited by terminal sampling in many observed profiles.

Within-Target Comparisons

Within-target comparisons included two antibodies each for PD-L1, PD-1, and TNF α . For PD-L1, human C_{max} predictions met the two-fold criterion for both atezolizumab and avelumab at all matched doses. $AUC_{0-tlast}$ predictions met the criterion for all atezolizumab doses, whereas the lowest avelumab dose did not. At the shared monkey dose, C_{max} met the criterion for both antibodies, whereas avelumab $AUC_{0-tlast}$ did not. For PD-1 in humans, all $AUC_{0-tlast}$ predictions and all but one C_{max} prediction for nivolumab and pembrolizumab met the two-fold criterion, including both endpoints at the matched dose. Monkey results were less consistent, with one $AUC_{0-tlast}$ prediction for nivolumab and one C_{max} prediction for pembrolizumab outside the criterion. For TNF α , all human $AUC_{0-tlast}$ and C_{max} predictions for adalimumab and golimumab met the two-fold criterion, including at matched doses. Clearance predictions also met the criterion at matched doses where this endpoint was evaluable for both antibodies. At the shared monkey dose, $AUC_{0-tlast}$ and C_{max} predictions met the criterion for both molecules. Because target reference concentration and target degradation rate were shared by antibodies against the same target, the within-target differences cannot be attributed to those shared inputs alone. The PD-L1 divergence is more consistent with molecule-specific FcRn binding, target-binding kinetics, internalization, or unmodeled antibody properties; the distinct PD-1 miss patterns likewise point to drug-specific inputs or data context. Conversely, the consistent TNF α results occurred with shared target-system inputs, complex internalization, and the common FcRn Kd, despite different target-binding kinetics. These comparisons generate mechanistic hypotheses rather than causal conclusions because each molecule was evaluated against different literature profiles and not all doses or populations were matched.

Evaluation of Simulation-Based Study Designs

Study design simulations were performed to determine whether smaller, sparsely sampled monkey studies could recover reference “true” PK endpoints under predefined assumptions about parameter uncertainty and between-subject variability. Golimumab and nivolumab represented two outcomes from the preceding predictive evaluation. The uncalibrated golimumab model adequately predicted the observed PK and was therefore used without modification. In contrast, the nivolumab model showed poorer agreement for AUC_{0–tlast} and clearance. To establish a representative truth model for the study-design analysis, FcRn binding affinity was fitted to observed mean PK data from two sparsely sampled dose groups. The original anti-PD-1 model structure and all other parameters were retained. These cases therefore illustrate proceeding with either an unmodified model when predictive performance is adequate or a targeted model refinement when it is not (Figure 3).

For each molecule and variability scenario (30% or 60% CV), 100 Monte Carlo replicates were generated. Positive-valued uncertain parameters were sampled from lognormal distributions centered on the corresponding truth-model values. Within each replicate, FcRn binding affinity, target binding affinity, target dissociation rate, drug–target complex internalization rate, and target degradation rate were drawn once and shared across all 50 monkeys and five doses. Between-monkey variability was limited to body weight and target reference concentration; endogenous IgG and FcRn abundance and function were held constant. Body weights were generated using the configured 3–7 kg PK-Sim population range. The target-concentration population was generated once for each molecule and variability scenario, recentered to the specified arithmetic mean, and reused across replicates and doses. Thus, target concentration varied among monkeys but was not independently resampled at the replicate level. Combining two molecules, two variability scenarios, 100 replicates, 50 monkeys, and five doses (0.1, 1, 3, 10, and 30 mg/kg) produced 100,000 population profiles.

Candidate studies included two dose groups (1 and 10 mg/kg), three dose groups (0.1, 1, and 10 mg/kg), or all five dose groups, with two, three, or six distinct monkeys per group. Dense profiles were sampled every 0.25 h through 8 h and hourly thereafter through day 28. Sparse sampling was performed at 0.25, 1, 2, 4, and 8 h and on days 1, 2, 6, 10, 14, 21, and 28. For each molecule, variability scenario, replicate, dose, and group size, monkeys were selected without replacement within a dose group and independently across dose groups. The same selected animals were used to calculate all endpoints, and designs sharing a dose and group size used the same animals for that dose.

The evaluated endpoints were AUC0–28d, Cmax, and clearance. AUC0–28d was calculated using the linear-up/log-down trapezoidal method, and Cmax was the maximum sampled concentration. The terminal phase was selected as the post-Cmax sequence of at least three points with the highest adjusted R^2 ; this phase was used to estimate AUCinf and clearance. When no time-zero sample was available, the first measured concentration was carried back to time zero.

For the primary analysis, fold error was calculated as the arithmetic mean of the sparse study-group endpoints divided by the arithmetic mean of the corresponding 50 replicate-matched dense individual endpoints. This comparison isolates the ability of a candidate design to recover the PK of its underlying simulated population. A secondary analysis instead used the fixed dense nominal typical-subject endpoint as the denominator, thereby incorporating both study-sampling error and the replicate-level shift from the nominal model.

Joint success required AUC0–28d, Cmax, and clearance at every included dose to fall within the prespecified 0.5- to 2-fold interval. Endpoint-specific success required the indicated endpoint to meet this criterion at every included dose. The two-fold interval was used as a tutorial performance criterion. The complete workflow and executable code are provided in the 03_Study_Simulations folder of Supplementary Material 1.

Study Design Simulation Outputs

In the primary replicate-matched analysis, all 36 combinations of molecule, variability scenario, dose-group design, and group size achieved 100% joint success across 100 replicates. This included the smallest design—1 and 10 mg/kg with two monkeys per group (total n=4)—for both molecules and both variability scenarios. Individual fold errors ranged from 0.624 to 1.386.

The secondary analysis, which used fixed nominal typical-subject endpoints as the reference, showed larger deviations because it captured both sparse-study sampling error and replicate-level shifts from the nominal model (**Tables 1 and 2**). Across designs, mean RMSE ranged from 26.7% to 29.6% for AUC0–28d, 13.1% to 17.6% for Cmax, and 34.5% to 34.6% for clearance. Endpoint-specific success ranged from 89.3% to 96.5% for AUC0–28d and from 93.0% to 93.8% for clearance, while Cmax success was 100% for all designs.

Together, these analyses indicate that two dose groups were sufficient for the objective of recovering PK endpoints under the simulated conditions. In the primary analysis, relative to the two-dose design, the three-dose design improved mean AUC0–28d and Cmax RMSE by no more than 0.16% , while the five-dose design reduced RMSE by at most 0.30%. With two dose groups, increasing the group size from two to three monkeys reduced AUC0–28d and

C_{max} RMSE from approximately 15.6% to 12.5%, while increasing from three to six monkeys per group further reduced RMSE to approximately 8.9% and 8.8%, respectively, with essentially no improvement in clearance. The secondary analysis likewise showed only modest reductions with six monkeys per group, particularly for AUC_{0–28d}, and no reduction in clearance RMSE because additional animals could not correct replicate-level parameter shifts. Thus, two monkeys per group represented the smallest adequate design evaluated (n=4), whereas three monkeys per group (n=6) provided a modest increase in precision and may offer a more conservative design choice. Increasing to six monkeys per group doubled animal use relative to three per group but provided limited additional practical benefit. These conclusions remain conditional on the truth models, variability assumptions, sampling schedule, endpoints, and success criterion examined.

Application and Key Considerations

This tutorial illustrates how an established computational framework, i.e., PBPK/TMDD models, can be used as a NAM to connect mechanistic input data with efforts to reduce the number of animals needed for preclinical assessments.

In this retrospective analysis, PBPK predictions generated without calibration to the analyzed in vivo PK data met the two-fold criterion for 85.7% of monkey and 96.7% of human AUC_{0–tlast} comparisons and for 92.9% of monkey and 96.7% of human C_{max} comparisons. Clearance was less consistently characterized: observed clearance was evaluable for only 6 of 14 monkey and 15 of 30 human scenarios, and 50.0% and 66.7% of those evaluable predictions, respectively, met the two-fold criterion. The high proportion of non-evaluable observed clearance estimates shows that late sampling and terminal-phase characterization materially constrain clearance-based model evaluation. Target reference concentration, target degradation rate, and complex internalization rate for several targets or molecules were derived from prior modeling analyses rather than molecule-specific in vitro measurements. Direct experimental characterization of these parameters may improve prospective prediction and would reduce dependence on assumptions inherited from earlier models. The local one-at-a-time sensitivity analysis in **Supplementary Material 2** supported this interpretation. Clearance was most responsive to target reference concentration and target degradation rate for tocilizumab in monkeys and humans, with smaller effects for selected bevacizumab and avelumab scenarios; complex internalization had a smaller effect across the evaluated cases. This analysis indicates that uncertainty in target-system inputs can contribute to selected clearance discrepancies, while the large number of non-evaluable observed clearances remains a separate consequence of limited terminal sampling.

The value of such approach depends on the quality and independence of the mechanistic inputs. Antibody-target binding affinity was available from experimental assays for all eight molecules, but the remaining inputs were less consistently measured (**Supplementary Material 2**). Molecule-specific FcRn binding affinity from SPR/Biacore was available for atezolizumab and pembrolizumab; for the other six mAbs, we used the human-derived default FcRn Kd of 1.07 $\mu\text{mol/L}$ obtained from a PBPK analysis of 50 mAbs [26]. The final 1.07- $\mu\text{mol/L}$ value was derived from the full 50-mAb dataset. Adalimumab, bevacizumab, and golimumab were included in that dataset and receive the default value here; adalimumab belonged to the original test subset, whereas bevacizumab and golimumab belonged to the training subset. Therefore, the human exposure evaluation for these three molecules is not fully independent of the data used to derive the default. Atezolizumab also appeared in the 50-mAb dataset but uses a molecule-specific measured FcRn value in this analysis. We also applied the human-derived 1.07- $\mu\text{mol/L}$ default to monkeys. In comparison, Stader et al. scaled human FcRn Kd by 0.45 for monkey simulations and reported strong overprediction of human exposure without an additional systemic clearance term [27]. We did not include an empirically fitted additional systemic clearance term; species scaling of FcRn affinity and unmodeled target-independent clearance may therefore contribute to residual errors. A measured complex internalization rate was available for only two molecules, while for others it was assumed to equal the target degradation rate. Target reference concentration was also difficult to standardize across targets and tissues. These substitutions are useful for demonstrating the workflow, but parameters estimated in earlier PK analyses may carry information from prior in vivo studies. Accordingly, prediction without calibration to the profiles evaluated here should not be interpreted as prediction from independent in vitro evidence alone. A prospective evaluation would benefit from a defined experimental package for binding, target abundance and localization, turnover, internalization, and target-independent clearance, with uncertainty ranges attached to each input.

In this example analysis, the study-design exercise addressed a narrower question: under specified levels of population variability and uncertainty in mechanistic inputs, could a smaller exploratory study recover the “true” PK characteristics? True PK is defined by the representative models that describe the observed data well in these analyses. Golimumab and nivolumab were selected from the preceding predictive evaluation to illustrate two outcomes within the PBPK-guided pathway shown in **Figure 3**. The uncalibrated golimumab model adequately predicted the observed PK and was therefore used without modification. In contrast, the nivolumab model showed poorer agreement for $\text{AUC}_{0-\text{last}}$ and clearance, so FcRn binding affinity alone was calibrated to observed mean PK data from only two sparsely sampled dose groups; the original anti-PD-1 model structure and all other parameters were retained. These examples illustrate proceeding with an unmodified

a priori model when its predictive performance is adequate or using limited exploratory in vivo data for targeted model refinement when it is not. In the primary replicate-matched analysis, every evaluated scenario met the joint two-fold performance criterion in 100/100 replicates, including the two-dose design with two monkeys per group for both molecules and variability levels. Four animals rather than 15 represents a 73% reduction relative to the modeled five-dose design with three animals per group. This is a comparison between the modeled designs, not a universal comparison with current industry practice or a reduction across an entire development program. The lower success rates in the secondary nominal-reference analysis show that reducing sampling error does not eliminate uncertainty in the underlying model inputs. **Figure 3** places these examples within a broader workflow, but the animal-number ranges are schematic. In practice, a sponsor may already use fewer dose groups, microsampling, cassette dosing, animal reuse, or PK/TK measurements incorporated into another study [17-18, 28]. Anti-drug antibodies may also eliminate late PK observations, while biological variability or additional pharmacodynamic and safety endpoints may require more animals. The model-informed design should therefore be compared with the program's actual feasible alternatives, including the use of data from an existing safety study or a suitable alternative model. The most defensible outcome is a fit-for-purpose design supported by explicit evidence and prespecified decision criteria.

The model is most useful for reducing study size when the intended decision is defined before the design is selected. The team should specify which exposure or PK features must be estimated, the required precision, and whether the study must also evaluate immunogenicity, target engagement, pharmacodynamics, or toxicity. PBPK simulations can then inform the selection of dose levels, dose groups, animal numbers, and sampling times needed to distinguish among plausible parameter sets. For a poorly characterized target, this process may indicate that a targeted in vitro measurement or a small pilot PK study would be more informative than proceeding directly to a larger, fixed design. Performance criteria should be defined in advance, and model inputs, data sources, and assumptions should be documented to support traceability and future refinement. Class-level linear mAb PK parameters can provide informative priors for early design simulations [29], while statistical planning should quantify uncertainty and avoid treating a small model-conditional exercise as universal evidence [30].

Our analysis focused on IV single-dose mAb PK, an appropriate initial context because the principal disposition processes are comparatively well characterized. Single-dose data and mechanistic models can support projection of repeat-dose exposure and accumulation, and published examples show successful translation for selected mAbs [28, 29, 31-32]. However, these examples do not establish that a dedicated repeat-dose PK study can be

omitted for every molecule. When scientifically and operationally appropriate, PK/TK sampling embedded in dose-range-finding or GLP repeat-dose toxicology studies may provide repeat-dose exposure information without a separate PK study [3028], but the toxicology studies and their animal use remain outside the scope of the present analysis. The framework could subsequently be extended to subcutaneous mAbs by incorporating lymphatic uptake, pre-systemic catabolism, bioavailability, and formulation- and injection-site-dependent absorption. Hybrid approaches using human FcRn transgenic mice are another potential comparator for human PK projection, but were not tested here. Lessons from mAbs may also inform PBPK or hybrid PBPK-QSP workflows for more complex modalities, ~~but~~ each extension will require its own context-of-use definition and validation. Related reviews and OSP-based translational analyses describe additional mechanistic and cross-species considerations for mAb PBPK modeling [33-34].

This study was a small retrospective proof-of-concept analysis constrained by the availability of public monkey PK, human PK, and mechanistic input data. All eight antibodies were marketed products against comparatively well-studied targets, so the results may not transfer to a first-in-class antibody for which target abundance, turnover, shedding, or localization is poorly characterized. Several target-system inputs, including target reference concentration, k_{int} , and k_{deg} , were taken from prior modeling studies rather than direct in vitro measurements. Because some ~~of those~~ values may have been selected with knowledge of ~~the~~ observed PK, the inputs are not fully independent and may make retrospective performance appear optimistic. Human PK for the oncology mAbs was obtained from patients with potentially variable and high tumor-target burden rather than first-in-human healthy volunteers; disease-related target expression and ~~patient-versus-healthy~~ population differences are ~~therefore~~ plausible contributors to prediction error. TheFor most source profiles, it was also unclear whether the assay measured free or total antibody, and some observed clearance estimates relied on substantial terminal extrapolation. Finally, the generic target representation ~~also~~ did not distinguish soluble from membrane target or explicitly represent shedding, despite differences among ~~soluble and transmembrane~~ TNF, ~~soluble~~ VEGF, ~~soluble and membrane~~ IL-6R, and ~~membrane~~ PD-1/PD-L1. ~~This level systems. These limitations restrict mechanistic interpretation and generalization of granularity was not required for both the initial context of use—evaluating exposure predictionpredictive results and the conditional precision of a reduced study-design—but it limits mechanistic interpretation and generalizability finding.~~

~~Alternative empirical TMDD approximations, including a Michaelis-Menten model, were not evaluated. Estimation of Vmax and Km together with the structural PK parameters would have been poorly supported for molecules represented by only one monkey dose level. Larger retrospective datasets and The next step is prospective pilot testing with a clearly~~

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defined context of use. Such studies should therefore generate obtain standardized in vitro mechanistic inputs where feasible, lock “a priori” models the PBPK model before observing in vivo examining new PK data, prespecify quantitative success criteria, include adequate and retain individual-level data, and evaluate target representations and model structures appropriate to each molecule. Such evidence is needed to define a defensible context of use and build regulatory confidence in PBPK-informed reduction of preclinical monkey PK studies.

PK-Sim/MoBi and the R packages OSP Suite and esqlabsR provided a practical open-source framework for a reproducible, semi-automated workflow [22-24]. The platform combines physiologically structured PBPK models with curated species-specific physiology and expression databases, allowing target expression profiles to be defined consistently across monkeys and humans [20, 21]. Export to MoBi and programmatic simulation in R also enable scalable evaluation across molecules, targets, species, and dose levels. This reproducibility is important for prospective PBPK-based NAM workflows, in which model generation, parameter updates, simulations, and exposure metric extraction should be prespecified and independently traceable.

Conclusions

This retrospective analysis supports further evaluation of “a priori” PBPK modeling as a NAM for first-in-human mAb PK prediction. Predictive performance of “a priori” approach for human PK predictions was broadly comparable to monkey-to-human top-down translation for the evaluated dataset, although clearance-related predictions were sensitive to uncertainty in target-system inputs as well as observed data. The reduced-study simulations support an illustrative, model-informed exploratory single-dose design of two dose groups with three animals, late samples, and immunogenicity information. A larger retrospective set, including multiple antibodies per group, corresponding to 6 rather than at least 15 animals (a 60% reduction) under the stated target and less characterized targets, would help distinguish molecule effects from target assumptions. This finding supports prospective validation of “a priori” Together, those data would show when PBPK modeling and can support a reduced exploratory study, when a different design is warranted, and how much confidence can be placed in human exposure projections.

Conclusions

This tutorial provides a reproducible method for using PBPK models to predict monkey and human mAb PK without calibration with in vivo PK profiles and to explore smaller monkey PK study designs as a NAM for first-in-human PK predictions. In the eight-antibody

retrospective example, the PBPK framework predicted both $AUC_{0-\text{last}}$ and C_{max} within two-fold of observed values across all evaluated doses for five of the eight molecules. Study-design simulations indicated that a two-dose group study with two animals (n=4 total) was sufficient to recover the model-defined PK endpoints, while two-dose group study with three animals each (n=6 total) provided modestly greater precision and can be considered a sufficient design. Prospective evaluation should use larger datasets, experimentally measured mechanistic inputs, prespecified decision criteria, and explicit assessment of study objectives, uncertainties, and model and study-design assumptions.

References

1. Sun D, Gao W, Hu H, Zhou S. Why 90% of clinical drug development fails and how to improve it?. *Acta Pharmaceutica Sinica B*. 2022;12(7):3049-3062. doi:10.1016/j.apsb.2022.02.002.
2. Garg A, Balthasar JP. Physiologically-based pharmacokinetic (PBPK) model to predict IgG tissue kinetics in wild-type and FcRn-knockout mice. *Journal of Pharmacokinetics and Pharmacodynamics*. 2007;34(5):687-709. doi:10.1007/s10928-007-9065-1.
3. Shah DK, Betts A. Towards a platform PBPK model to characterize the plasma and tissue disposition of monoclonal antibodies in preclinical species and human. *Journal of Pharmacokinetics and Pharmacodynamics*. 2011;39(1):67-86. doi:10.1007/s10928-011-9232-2.
4. Niederalt C, Kuepfer L, Solodenko J, Eißing T, Siegmund H, Block MS, et al. A generic whole body physiologically based pharmacokinetic model for therapeutic proteins in PK-Sim. *Journal of Pharmacokinetics and Pharmacodynamics*. 2018;45(2):235-257. doi:10.1007/s10928-017-9559-4.
5. Jones HM, Zhang Z, Jasper P, Luo H, Avery LB, King LE, et al. [A Physiologically-Based Pharmacokinetic Model for the Prediction of Monoclonal Antibody Pharmacokinetics From In Vitro Data](#). *CPT: Pharmacometrics & Systems Pharmacology*. 2019;8(10):738-747. doi:10.1002/psp4.12461.
6. Hu S, Datta-Mannan A, D'Argenio DZ. [Physiologically Based Modeling to Predict Monoclonal Antibody Pharmacokinetics in Humans from In Vitro Physiochemical Properties](#). *mAbs*. 2022;14(1):2056944. doi:10.1080/19420862.2022.2056944.
7. US Food and Drug Administration. Roadmap to Reducing Animal Testing in Preclinical Safety Studies. Silver Spring, MD: FDA; April 2025. Available from: <https://www.fda.gov/media/186092/download> (accessed July 26, 2026).

68. US Food and Drug Administration. General Considerations for the Use of New Approach Methodologies in Drug Development. Draft Guidance for Industry. March 2026. Docket FDA-2025-D-6131. Available from: <https://www.fda.gov/regulatory-information/search-fda-guidance-documents/general-considerations-use-new-approach-methodologies-drug-development> (accessed July 26, 2026).

79. US Food and Drug Administration. Reducing Animal Testing in Nonclinical Studies: Year One Progress and the Path Forward. April 2026. Available from: <https://www.fda.gov/media/191986/download> (accessed July 26, 2026).

810. Mehta K, Maaß C, Cucurull-Sanchez L, Pichardo-Almarza C, Subramanian K, Androulakis IP, et al. Modernizing Preclinical Drug Development: The Role of New Approach Methodologies. ACS Pharmacology & Translational Science. 2025;8(6):1513-1525. doi:10.1021/acspsci.5c00162.

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911. National Institutes of Health Common Fund. NIH invests \$150 million in human-based research to reduce use of animal models. March 18, 2026. Available from: <https://commonfund.nih.gov/complementarie/news/nih-invests-150-million-human-based-research-reduce-use-animal-models> (accessed July 26, 2026).

Formatted: German (Germany)

1012. Sillé FC, Härtung T. Metabolomics in Preclinical Drug Safety Assessment: Current Status and Future Trends. Metabolites. 2024;14(2):98. doi:10.3390/metabo14020098.

Formatted: English (United States)

1113. Kreutz A, Chang X, Högberg HT, Wetmore BA. Advancing understanding of human variability through toxicokinetic modeling, in vitro-in vivo extrapolation, and new approach methodologies. Human Genomics. 2024;18(1):129. doi:10.1186/s40246-024-00691-9.

Formatted: English (United States)

1214. Dao T, Sadrieh N. A CDER perspective: Landscape of New Approach Methodologies (NAMs) submitted in drug development programs. Regulatory Toxicology and Pharmacology. 2025;165:106007. doi:10.1016/j.yrtph.2025.106007.

1315. Bai JPF, Liu G, Zhao M, Wang J, Xiong Y, Truong TM, et al. Landscape of regulatory quantitative systems pharmacology submissions to the U.S. Food and Drug Administration: An update report. CPT: Pharmacometrics & Systems Pharmacology. 2024;13(12):2102-2110. doi:10.1002/psp4.13208.

14. Jones HM, Zhang Z, Jasper P, Luo H, Avery LB, King LE, et al. A Physiologically-Based Pharmacokinetic Model for the Prediction of Monoclonal Antibody Pharmacokinetics From In Vitro Data. CPT Pharmacometrics & Systems Pharmacology. 2019;8(10):738-747. doi:10.1002/psp4.12461.

~~15. Hu S, Datta-Mannan A, D'Argenio DZ. Physiologically Based Modeling to Predict Monoclonal Antibody Pharmacokinetics in Humans from in vitro Physiochemical Properties. mAbs. 2022;14(1):2056944. doi:10.1080/19420862.2022.2056944.~~

16. Germovsek E, Cheng M, Giragossian C. Allometric Scaling of Therapeutic Monoclonal Antibodies in Preclinical and Clinical Settings. *mAbs*. 2021;13(1):1964935. doi:10.1080/19420862.2021.1964935.

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17. Ménochet K, Yu H, Wang B, Tibbitts J, Hsu CP, Kamath AV, Richter WF, Baumann A. Non-human Primates in the PKPD Evaluation of Biologics: Needs and Options to Reduce, Refine, and Replace. A BioSafe White Paper. *mAbs*. 2022;14(1):2145997. doi:10.1080/19420862.2022.2145997.

Formatted: Font: Not Italic

18. Nagayasu M, Ozeki K. Combination of Cassette-Dosing and Microsampling for Reduced Animal Usage for Antibody Pharmacokinetics in Cynomolgus Monkeys, Wild-Type Mice, and Human FcRn Transgenic Mice. *Pharmaceutical Research*. 2021;38(4):583–592. doi:10.1007/s11095-021-03028-6.

Formatted: Font: Not Italic

~~19.19. Lippert J, Burghaus R, Edginton AN, Frechen S, Karlsson MO, Kovar A, et al. Open Systems Pharmacology Community—An Open Access, Open Source, Open Science Approach to Modeling and Simulation in Pharmaceutical Sciences. *CPT: Pharmacometrics & Systems Pharmacology*. 2019;8(12):878-882. doi:10.1002/psp4.12473.~~

20. esqLABS GmbH. esqlabsR: Utility Functions for Modelling and Simulation Workflows within ESQlabs Organization. R package version 5.6.0. Released March 31, 2026. Available from: <https://github.com/esqLABS/esqlabsR/releases/tag/v5.6.0> (accessed July 26, 2026).

21. Open Systems Pharmacology. ospsuite: R Interface to the Open Systems Pharmacology Suite. R package version 12.4.2. Released April 8, 2026. Available from: <https://github.com/Open-Systems-Pharmacology/OSPSuite-R/releases/tag/v12.4.2> (accessed July 26, 2026).

22. Meyer M, Schneckener S, Ludewig B, Kuepfer L, Lippert J. Using Expression Data for Quantification of Active Processes in Physiologically Based Pharmacokinetic Modeling. *Drug Metabolism and Disposition*. 2012;40(5):892-901. doi:10.1124/dmd.111.043174.

23. Cordes H, Rapp H. Gene expression databases for physiologically based pharmacokinetic modeling of humans and animal species. *CPT: Pharmacometrics & Systems Pharmacology*. 2023;12(3):311-319. doi:10.1002/psp4.12904.

24. Geci R, Gadaleta D, García de Lomana M, Ortega-Vallbona R, Colombo E, Serrano-Candelas E, et al. Systematic evaluation of high-throughput PBK modelling strategies for

[the prediction of intravenous and oral pharmacokinetics in humans. Archives of Toxicology. 2024;98\(8\):2659-2676. doi:10.1007/s00204-024-03764-9.](#)

25. Open Systems Pharmacology. Modularization concept. Open Systems Pharmacology Suite documentation, version 12. Available from: <https://docs.open-systems-pharmacology.org/v12/working-with-mobi/mobi-documentation/modularization-concept> (accessed July 26, 2026).

26. Benamara S, Sjögren E, Gattacceca F, Chenel M, Deslandes A, Nguyen L, Teutonico D. Prediction of Monoclonal Antibodies Pharmacokinetics in Human: Identification of a Reference Neonatal Fc Receptor (FcRn) Binding Affinity Using Physiologically Based Pharmacokinetic (PBPK) Modeling. *ACS Pharmacology & Translational Science*. 2026;9(1):177-190. doi:10.1021/acsptsci.5c00674.

27. Stader F, Sharma P, Huang W, Choules MP, Willemin ME, Zhang X, et al. Physiologically Based Pharmacokinetic Modeling of Biologic Case Studies in Monkeys and Humans Reveals the Necessity of an Additional Clearance Term. *Pharmaceutics*. 2025;17(5):560. doi:10.3390/pharmaceutics17050560.

28. Chapman K, Holzgrefe HH, Black LE, Brown MJ, Chellman GJ, Copeman C, et al. Pharmaceutical toxicology: Designing studies to reduce animal use, while maximizing human translation. *Regulatory Toxicology and Pharmacology*. 2013;66(1):88-103. doi:10.1016/j.yrtph.2013.03.001.

20. Meyer M, Schneckenor S, Ludewig B, Kuepfer L, Lippert J. Using Expression Data for Quantification of Active Processes in Physiologically Based Pharmacokinetic Modeling. *Drug Metabolism and Disposition*. 2012;40(5):992-991. doi:10.1124/dmd.111.043174.

21. Cordes H, Rapp H. Gene expression databases for physiologically based pharmacokinetic modeling of humans and animal species. *CPT Pharmacometrics & Systems Pharmacology*. 2023;12(3):311-319. doi:10.1002/psp4.12904.

22. Lippert J, Burghaus R, Edginton AN, Frechen S, Karlsson MG, Kovar A, et al. Open Systems Pharmacology Community: An Open Access, Open Source, Open Science Approach to Modeling and Simulation in Pharmaceutical Sciences. *CPT Pharmacometrics & Systems Pharmacology*. 2019;8(12):870-882. doi:10.1002/psp4.12473.

23. esqLABS GmbH. esqlabsR: Utility Functions for Modelling and Simulation Workflows within ESQlabs Organization. R package version 5.6.0. Released March 31, 2026. Available from: <https://github.com/esqLABS/esqlabsR/releases/tag/v5.6.0> (accessed July 26, 2026).

24. Open Systems Pharmacology. OSP Suite: R Interface to the Open Systems Pharmacology Suite. R package version 12.4.2. Released April 8, 2026. Available from:

~~<https://github.com/Open-Systems-Pharmacology/OSPSuite-R/releases/tag/v12.4.2>~~
(accessed July 26, 2026).

2529. Betts A, Keunecke A, van Steeg TJ, van der Graaf PH, Avery LB, Jones H, Berkhout J. Linear pharmacokinetic parameters for monoclonal antibodies are similar within a species and across different pharmacological targets: A comparison between human, cynomolgus monkey and hFcRn Tg32 transgenic mouse using a population-modeling approach. *mAbs*. 2018;10(5):751-764. doi:10.1080/19420862.2018.1462429.

2630. Gosselin RD. Guidelines on statistics for researchers using laboratory animals: the essentials. *Laboratory Animals*. 2019;53(1):28-42. doi:10.1177/0023677218783223.

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~~2731. Benamara S, Sjögren E, Gattacceca F, Chenet M, Deslandes A, Nguyen L, Teutonico D. Prediction of Monoclonal Antibodies Pharmacokinetics in Human: Identification of a Reference Neonatal Fc Receptor (FcRn) Binding Affinity Using Physiologically Based Pharmacokinetic (PBPK) Modeling. *ACS Pharmacology & Translational Science*. 2026;9:177-190. doi:10.1021/acspsci.5c00674.~~

28. Deng R, Iyer S, Theil F, Mortensen DL, Fielder PJ, Prabhu S. Projecting human pharmacokinetics of therapeutic antibodies from nonclinical data. *mAbs*. 2011;3(1):61-66. doi:10.4161/mabs.3.1.13799.

2932. Dong J, Salinger DH, Endres CJ, Gibbs JP, Hsu C, Stouch B, et al. Quantitative Prediction of Human Pharmacokinetics for Monoclonal Antibodies. *Clinical Pharmacokinetics*. 2011;50(2):131-142. doi:10.2165/11537430-000000000-00000.

~~30. Chapman K, Holzgrove HH, Black LE, Brown MJ, Chellman GJ, Gopeman G, et al. Pharmaceutical toxicology: Designing studies to reduce animal use, while maximizing human translation. *Regulatory Toxicology and Pharmacology*. 2013;66(1):88-103. doi:10.1016/j.yrtph.2013.03.001.~~

33. Glassman PM, Balthasar JP. Physiologically-based modeling of monoclonal antibody pharmacokinetics in drug discovery and development. *Drug Metabolism and Pharmacokinetics*. 2019;34(1):3-13. doi:10.1016/j.dmpk.2018.11.002.

34. Chen W, Wang L, Ruan Z, Lou H, Jiang B. Translate Pharmacokinetics of PD-1/PD-L1 Monoclonal Antibodies from Cynomolgus Monkey to Human: Comparison of Different Approaches. *Journal of Pharmaceutical Sciences*. 2024;113(9):2915-2921. doi:10.1016/j.xphs.2024.07.003.

Figures

Figure 1. (A) Overview of the tutorial workflow.

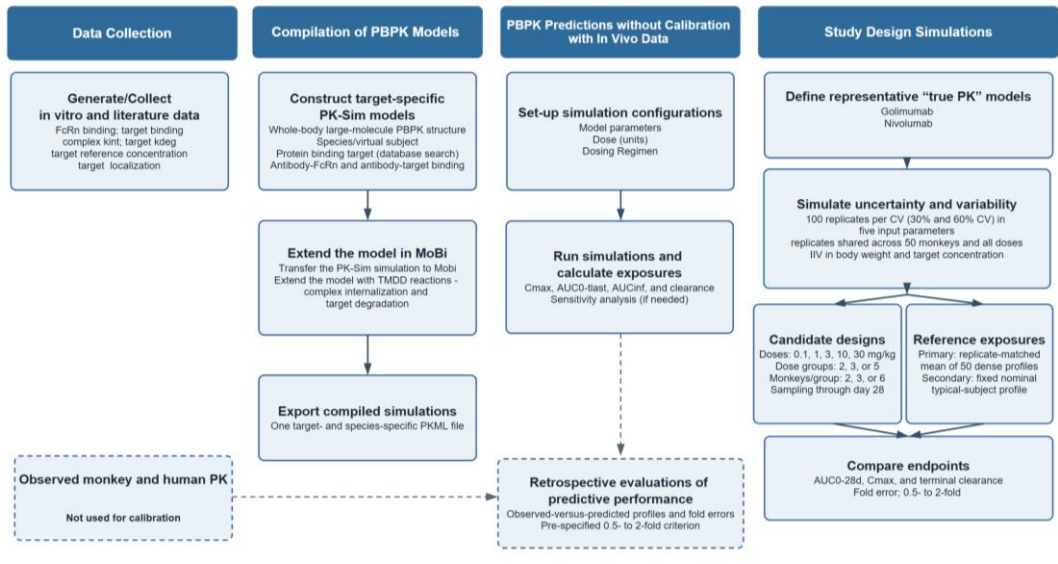
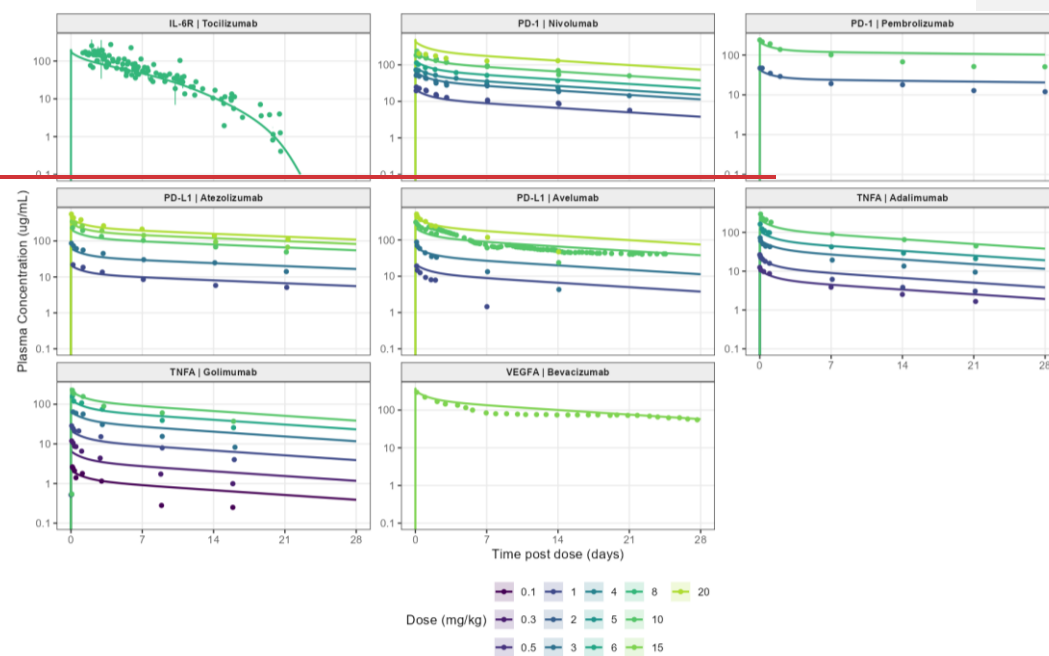


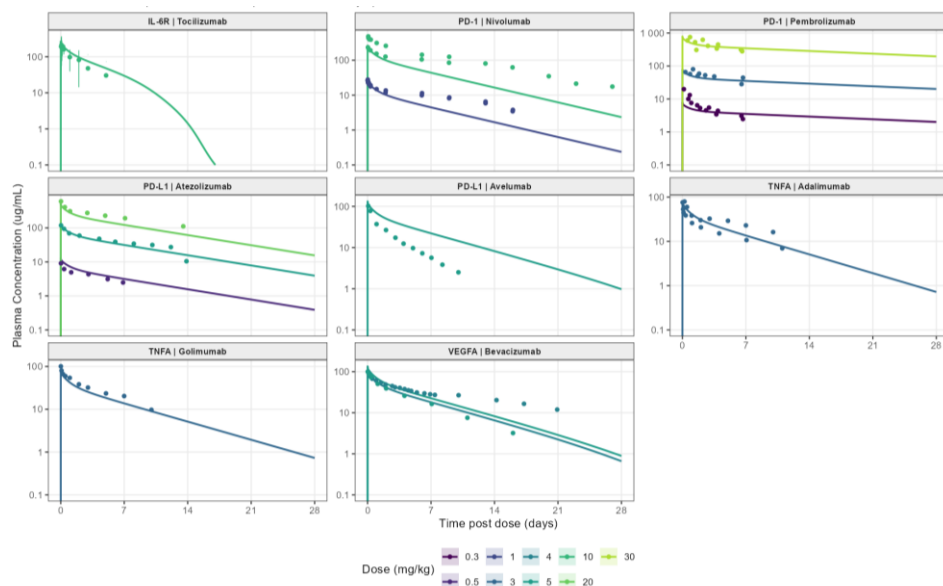
Figure 2. Observed and “a-priori” PBPK-predicted human plasma concentration-time profiles without calibration to the analyzed in vivo PK data across molecules and dose levels, (B) Observed and predicted human plasma concentration-time profiles from monkey-to-human translation of the top-down QSS-TMDD approach.

(A)



in (A) monkeys and (B) humans.

(A) Monkeys



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(B) Humans

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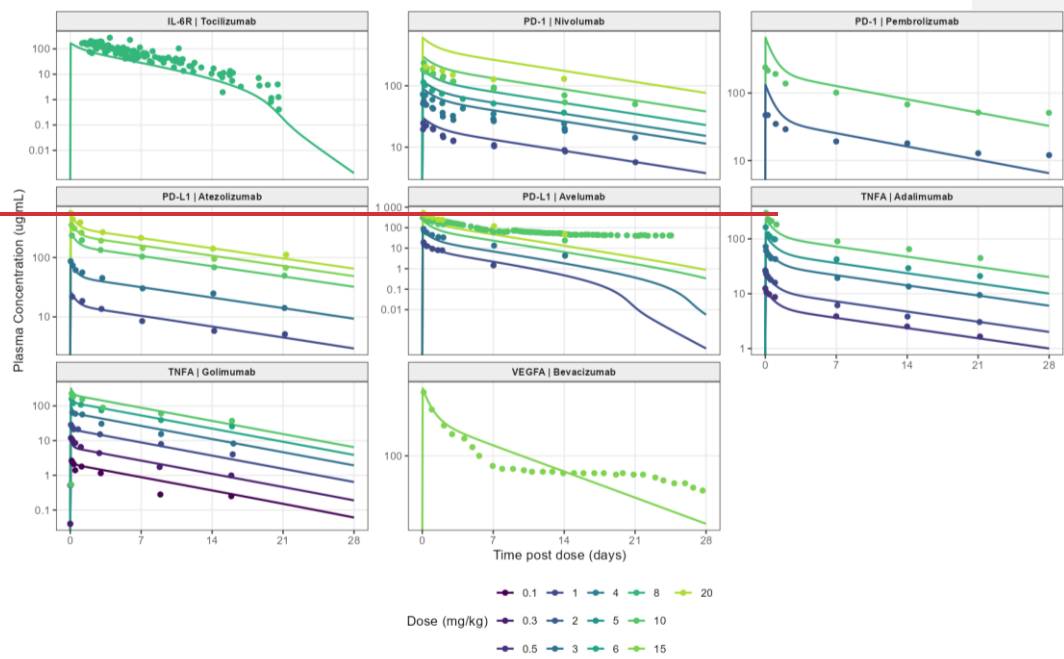
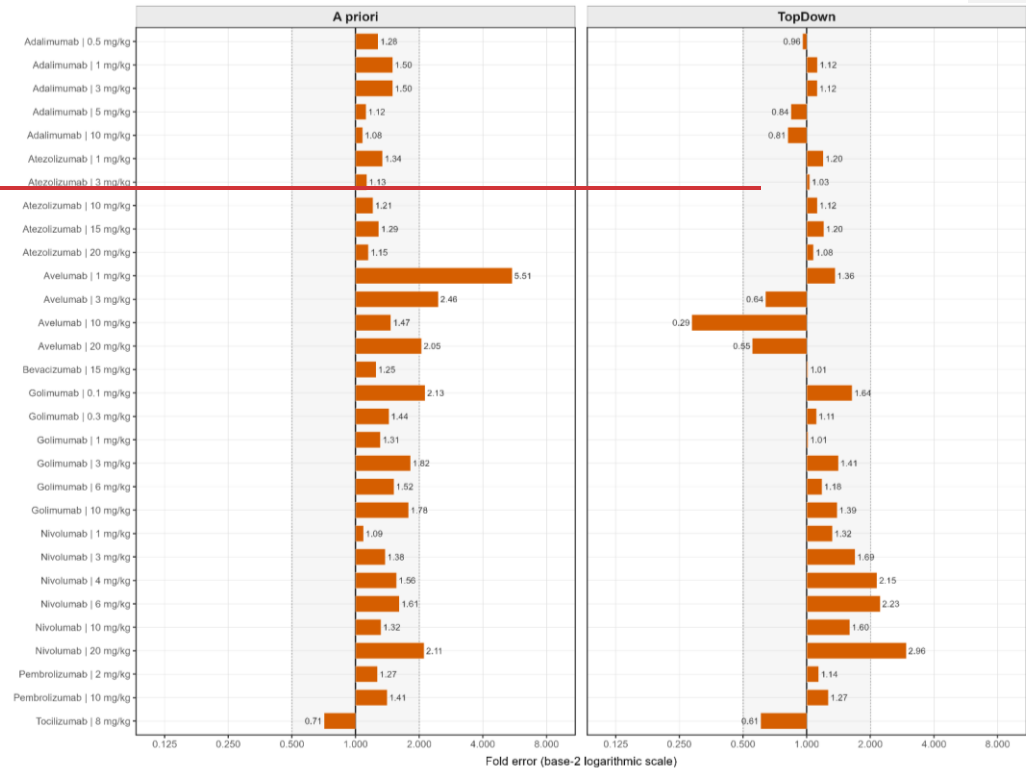
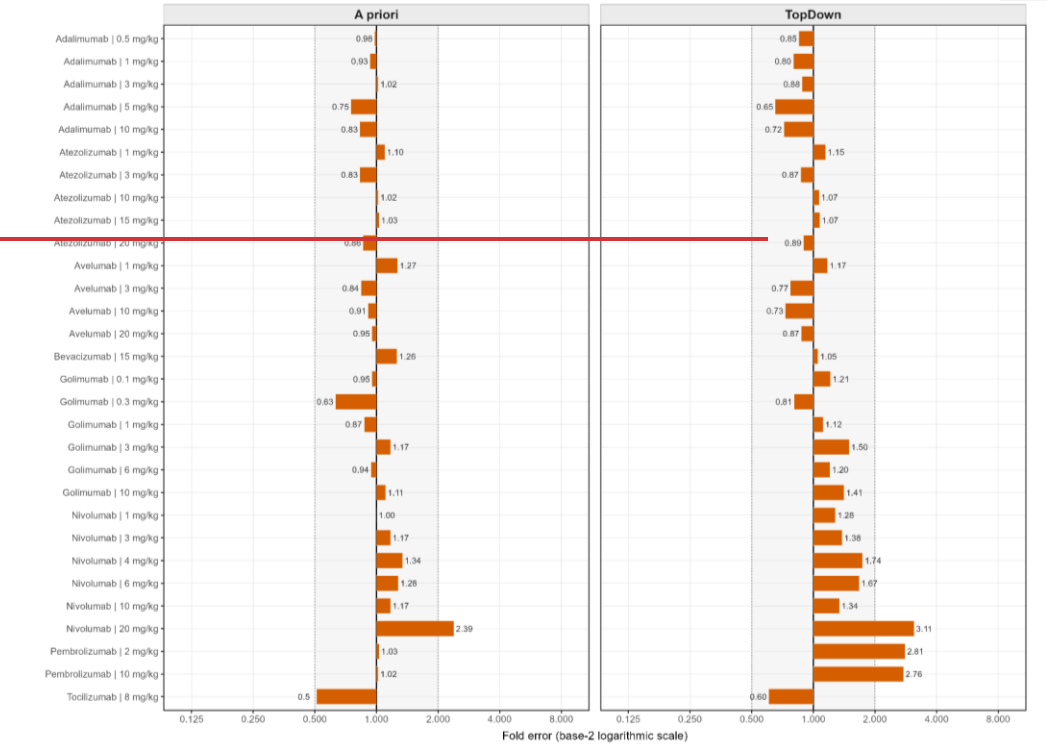


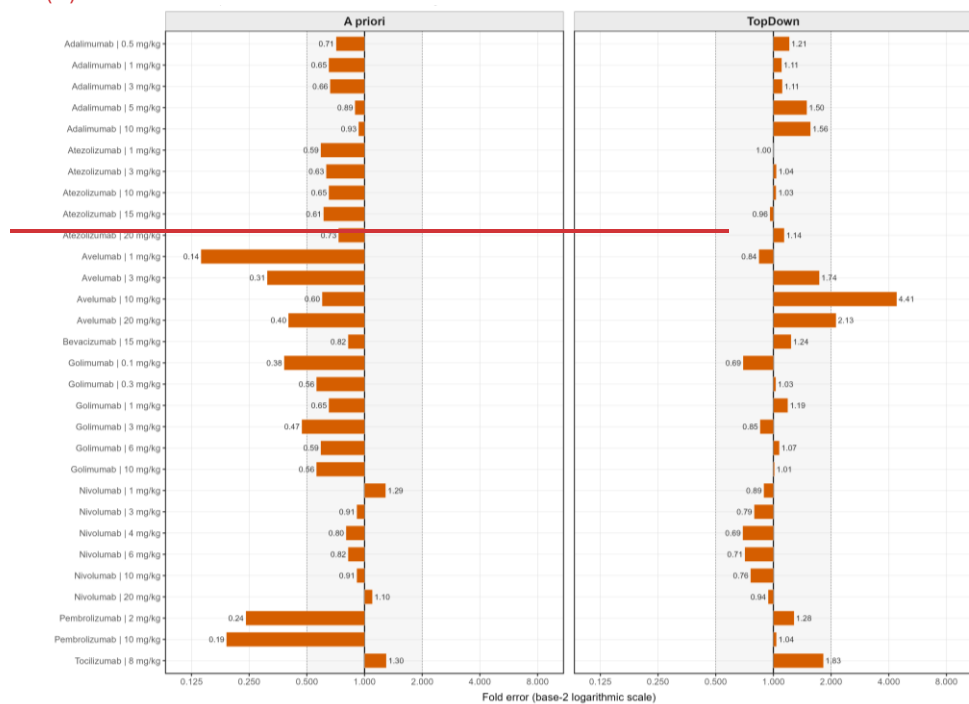
Figure 2. Fold-error comparison for human clearance, AUC0-28d, and Gmax predicted using “a priori” PBPK and monkey-fitted top-down TMDD translation. (A) AUC0-28d; (B) Gmax, and (C) Clearance, calculated as dose/AUCinf and representing total systemic clearance; At target-saturating doses, clearance is expected to be dominated by linear clearance. Fold error equals predicted divided by observed; bars extend from 1 on a base-2 logarithmic scale, and the shaded region and dashed lines indicate the 0.5- to 2-fold interval.

(A)



(B)





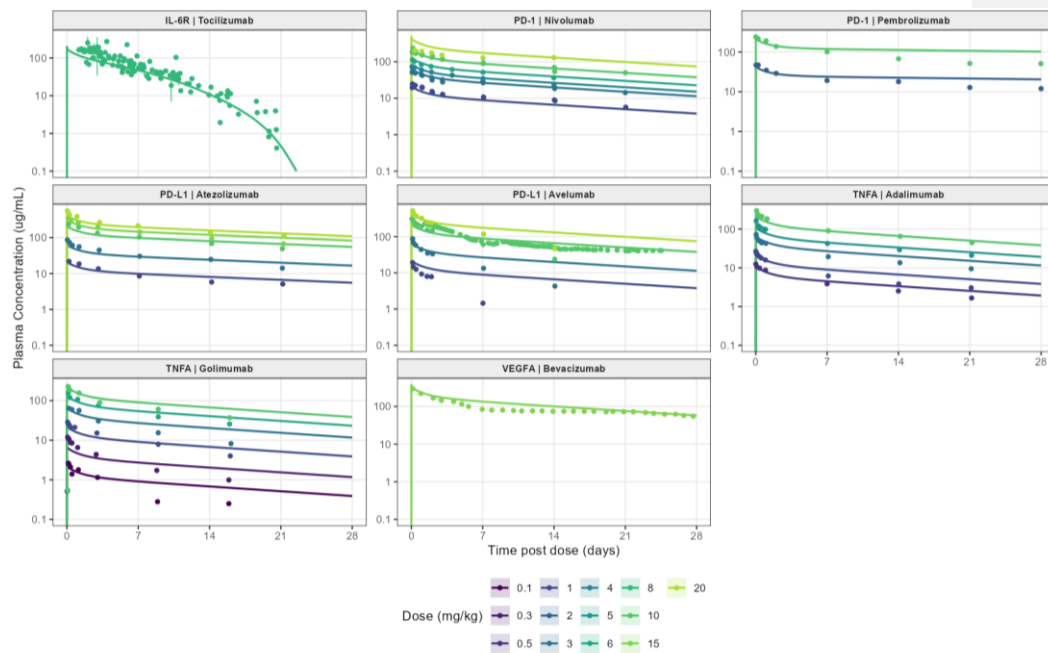


Figure 3. Model-conditional fold-error distributions for reduced monkey PK study designs. Fold error is the reduced-study arithmetic mean divided by the richly-sampled top-down population arithmetic mean. The primary analysis applies 30% interindividual variability to CL, V1, Q, and V2; fixed TMDD parameters are unchanged. The shaded interval indicates the prespecified 0.5- to 2-fold criterion.

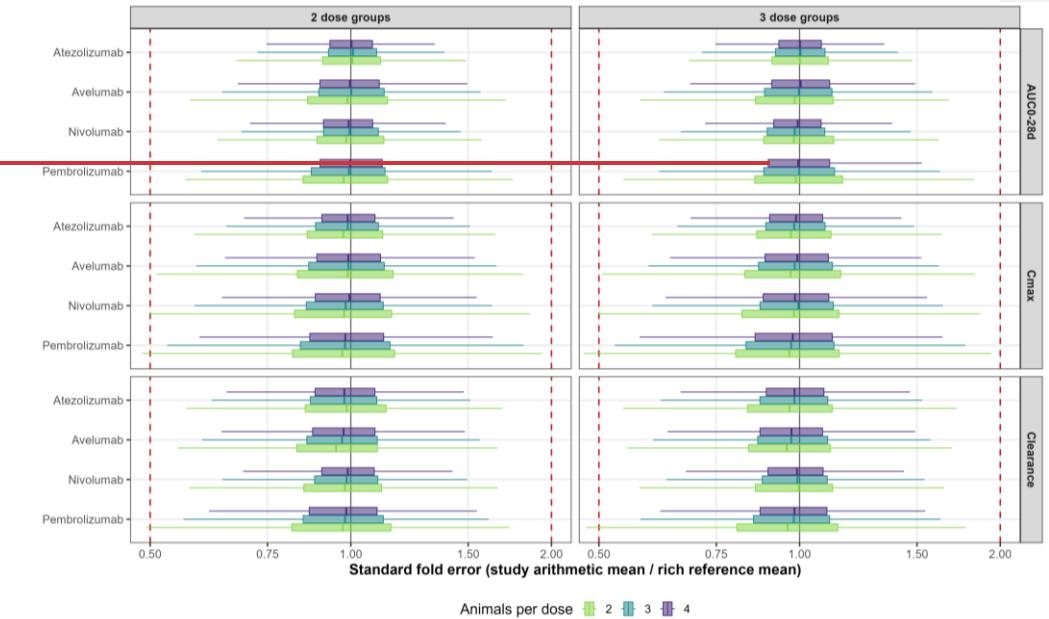
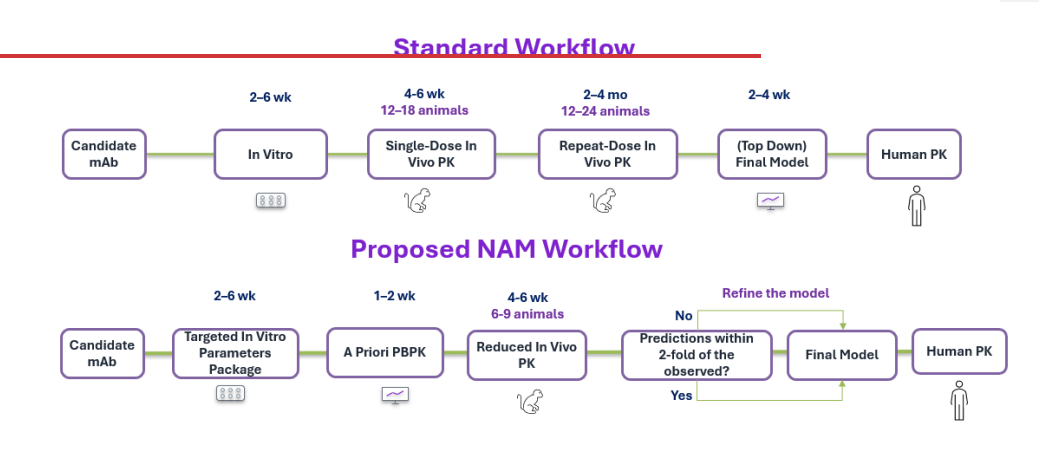


Figure 4. Proposed “a priori” illustrative PBPK-driven workflow for monoclonal antibody PK translation as a NAM: characterization with fewer animals. Molecule- and target-specific in vitro data mechanistic inputs are integrated into “a priori” PBPK models before in vivo PK testing. When residual remaining uncertainty warrants an in vivo study, PBPK predictions can inform a targeted exploratory single-dose design, and the resulting data can support model refinement and human exposure prediction. The illustrated animal counts are schematic examples rather than industry benchmarks; the modeled reduction applies to the compared single-dose PK and does designs, not apply to toxicology studies or the overall development program. FIH = first-in-human; PBPK = physiologically based pharmacokinetics. Figure generated with GPT-5.5.

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Standard PK characterization workflow

Sequential in vivo studies followed by top-down model development



PBPK-guided PK characterization workflow

A priori predictions inform in vivo studies to support model refinement



Table 1. RMSE by study design using the nominal typical subject as reference.

Dose groups	Monkeys per group	Total animals	RMSE, mean (range), %		
			AUC0-28d	Cmax	Clearance
2	2	4	29.0 (19.1-36.9)	17.4 (15.4-18.6)	34.6 (19.3-51.0)
2	3	6	28.5 (16.9-37.9)	15.5 (13.6-17.1)	34.6 (19.3-51.1)
2	6	12	26.9 (16.6-37.2)	13.1 (12.0-14.0)	34.6 (19.3-51.1)
3	2	6	29.6 (19.1-41.2)	17.5 (15.4-18.6)	34.5 (19.2-51.0)
3	3	9	28.6 (16.9-37.9)	15.6 (13.6-17.1)	34.5 (19.2-51.1)
3	6	18	26.7 (16.5-37.2)	13.2 (12.0-14.0)	34.5 (19.2-51.1)
5	2	10	29.5 (19.1-41.2)	17.6 (15.4-19.5)	34.6 (19.2-51.6)
5	3	15	28.3 (16.9-37.9)	15.4 (13.6-17.6)	34.6 (19.2-51.7)
5	6	30	26.8 (16.5-39.0)	13.2 (12.0-14.3)	34.6 (19.2-51.5)

Note: RMSE was calculated across 100 Monte Carlo replicates within each molecule, variability, dose, design, group-size, and endpoint cell. Entries show the arithmetic mean and range across both molecules, both variability scenarios, and all included doses.

Table 2. Replicates within two-fold by study design using the nominal typical subject as reference.

Dose groups	Monkeys per group	Total animals	Replicates with all included doses within 2-fold, %		
			AUC0-28d	Cmax	Clearance
2	2	4	92.3	100.0	93.5
2	3	6	95.3	100.0	93.8
2	6	12	96.5	100.0	93.8
3	2	6	91.0	100.0	93.5
3	3	9	94.5	100.0	93.8
3	6	18	95.8	100.0	93.8
5	2	10	91.3	100.0	93.0
5	3	15	93.0	100.0	93.5
5	6	30	94.0	100.0	93.5

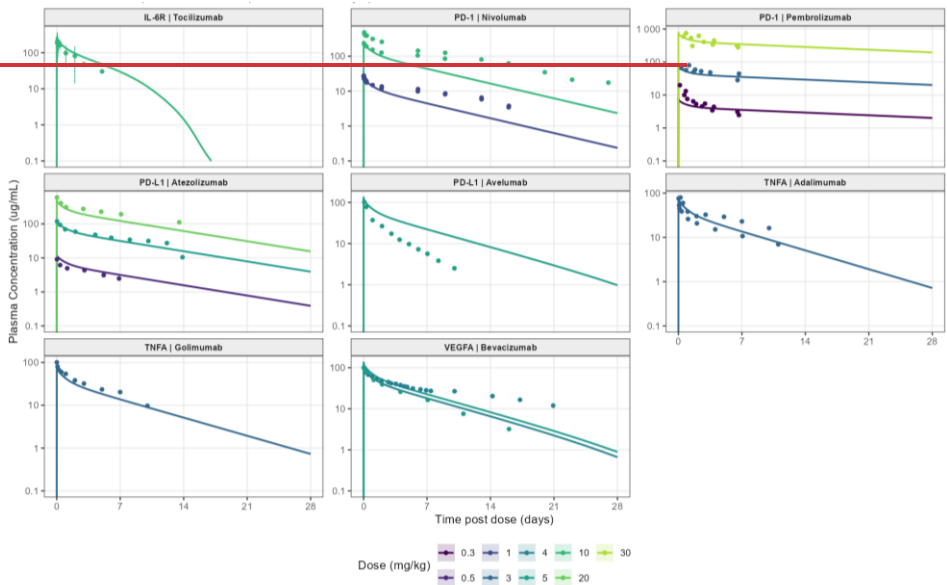
Note: Success percentages pool 400 parameter-draw replicates per row (2 molecules x 2 variability scenarios x 100 replicates). For each endpoint, a replicate was successful only when that endpoint was within 0.5-2.0 at every included dose.

Supplementary Materials

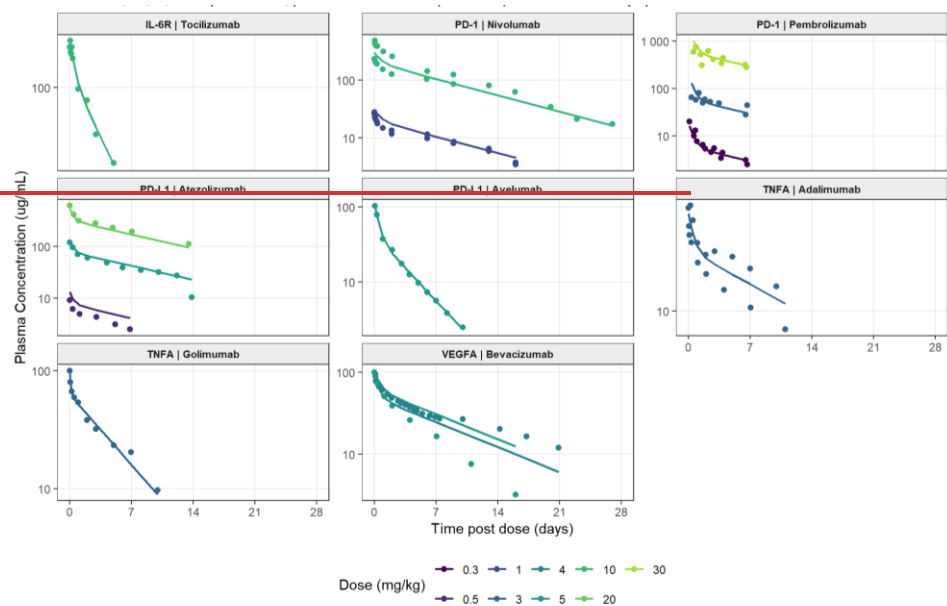
Supplementary Material S1. (S1.1) Observed and predicted monkey plasma concentration-time profiles for the “a priori” PBPK approach, and (B) the top-down QSS TMDD approach, (S1.2) Fold-error comparison for monkey clearance, AUC0-28d, and Gmax-predicted using “a priori” PBPK and monkey-fitted top-down TMDD translation, (S1.3) Parameters Used in A-Priori and Top-Down Simulations

S1.1. Observed and model-predicted monkey plasma concentration-time profiles across molecules and dose levels. (A) A priori PBPK simulations generated without fitting to monkey PK data. (B) Top-down QSS TMDD Run 4 fits with CL, Vc, Q, and Vp estimated and molecule-specific TMDD inputs fixed. Points represent observed data and lines represent model predictions.

(A) A priori PBPK monkey profiles

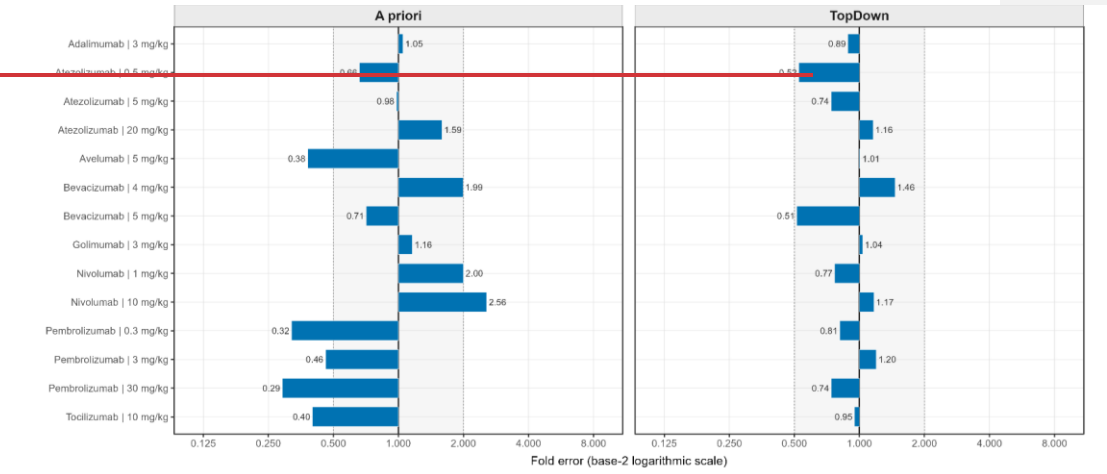


(B) Top-down-QSS-TMDD-monkey-fitted-profiles

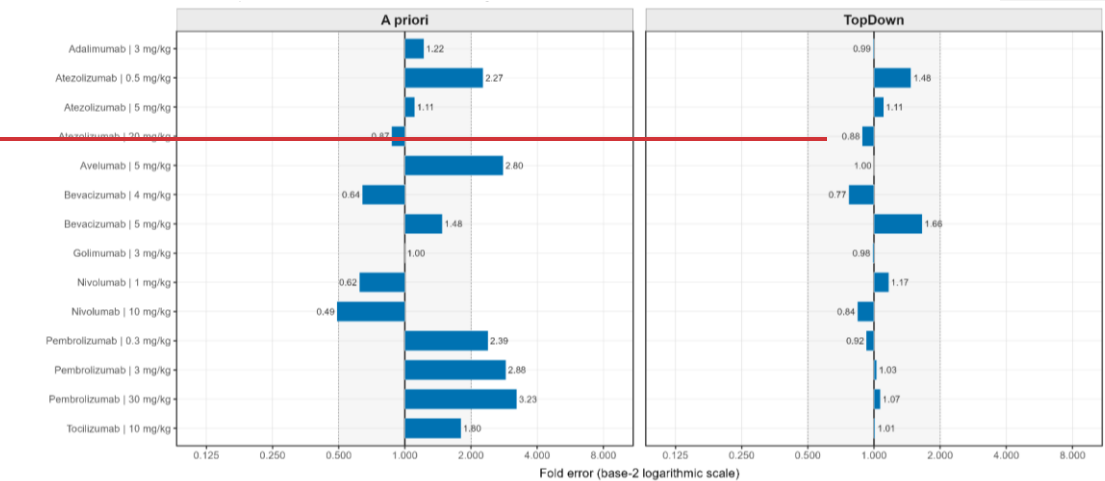


S1.2. Fold-error comparison for monkey clearance, AUC0-28d, and Gmax predicted using “a priori” PBPK and monkey-fitted top-down TMDD translation. (A) Clearance, calculated as dose/AUCinf and representing total systemic clearance; (B) AUC0-28d; and (C) Gmax. At target-saturating doses, clearance is expected to be dominated by linear clearance. Fold error equals predicted divided by observed; bars extend from 1 on a base-2 logarithmic scale, and the shaded region and dashed lines indicate the 0.5- to 2-fold interval.

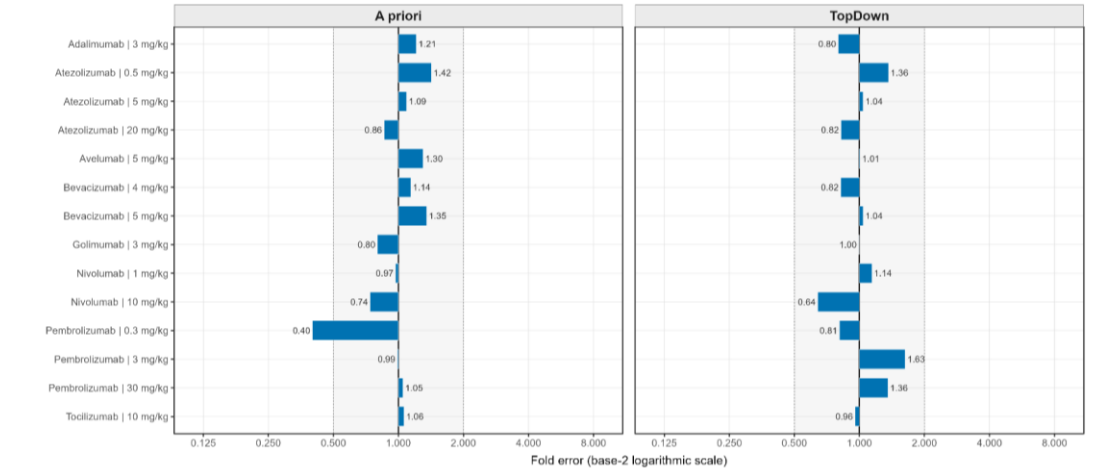
(A) Monkey clearance



(B) Monkey AUC0-28d



(C) Monkey Gmax



S1.3. Parameters Used in A Priori and Top-Down Simulations

Supplementary Material 1. GitHub repository containing the reproducible code (version tag v0.1.0):

https://github.com/namlink-org/PBPK_mAb_NAM_Tutorial

Supplementary Material 2. Additional-Figures-Tables.docx

S2.1. Parameters Used in the A Priori Predictions

Molecule	Species	Parameter	Value	Unit	Analytical Method for deriving the parameter	Parameter Group	Source
Adalimumab	Monkey	Kd	0.000127	uM	SPR/Biacore (Biacore 3000)	A-Priori and Top-Down	[1]
Adalimumab	Monkey	koff	0.0096	1/min	SPR/Biacore (Biacore 3000)	A-Priori and Top-Down	[1]
Adalimumab	Monkey	kint	0.866	1/h	Model fitting by the authors of [2]	A-Priori and Top-Down	[2]
Adalimumab	Monkey	Reference target concentration	0.001	nM	Model fitting by the authors of [3]	A-Priori and Top-Down	[3]
Adalimumab	Monkey	kdeg	0.223	1/h	Model parameter from anti-TNF PBPK analysis	A-Priori and Top-Down	[4]

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Adalimumab	Monkey	Kd (FcRn) in endosomal space	1.07	uM	Model-based meta-analysis [5]	A-Priori	[5]
Atezolizumab	Monkey	Kd	0.2990 00299	uM	Equilibrium binding studies using human PD-L1 expressed on 293 cells	A-Priori and Top Down	[6]
Atezolizumab	Monkey	koff	0.0093	1/min	SPR/Biacore	A-Priori and Top Down	[6]
Atezolizumab	Monkey	kint	0.606	1/h	Flow cytometry internalization assay in BxPC-3 cells	A-Priori and Top Down	[6]
Atezolizumab	Monkey	Reference target concentration	0.609	nM	Model fitting by the authors of [6]	A-Priori and Top Down	[6]
Atezolizumab	Monkey	kdeg	0.0426	1/h	Model fitting by the authors of [6]	A-Priori and Top Down	[6]
Atezolizumab	Monkey	Kd (FcRn) in endosomal space	0.65	uM	SPR/Biacore	A-Priori	[6]
Avelumab	Monkey	Kd	0.0000	uM	SPR assay (Biacore)	A-Priori and Top	[7]

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Avelumab	Monkey	koff	0.00174	1/min	SPR assay (Biacore T200)	A-Priori and Top Down	[8]
Avelumab	Monkey	kint	0.25	1/h	pHrodo flow-cytometry internalization assay	A-Priori and Top Down	[7]
Avelumab	Monkey	Reference target concentration	0.609	nM	Model fitting by the authors [6]	A-Priori and Top Down	[6]
Avelumab	Monkey	kdeg	0.0426	1/h	Model fitting by the authors [6]	A-Priori and Top Down	[6]
Avelumab	Monkey	Kd (FcRn) in endosomal space	1.07	uM	Model-based meta-analysis	A-Priori	[5]
Bevacizumab	Monkey	Kd	0.000058	uM	SPR-Biacore	A-Priori and Top Down	[9]
Bevacizumab	Monkey	koff	0.00186	1/min	SPR-Biacore	A-Priori and Top Down	[9]

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Bevacizumab	Monkey	kint	0.48	1/h	Assumed same as kdeg	A-Priori and Top Down	[10]
Bevacizumab	Monkey	Reference target concentration	0.013	nM	quantitative fluorescence	A-Priori and Top Down	[11]
Bevacizumab	Monkey	kdeg	0.48	1/h	cycloheximide-chase protein degradation assay; western blot dosimetry	A-Priori and Top Down	[10]
Bevacizumab	Monkey	Kd (FcRn) in endosomal space	1.07	uM	Model-based meta-analysis	A-Priori	[5]
Golimumab	Monkey	Kd	0.000018	uM	SPR/Biacore (Biacore 3000)	A-Priori and Top Down	[1]
Golimumab	Monkey	koff	0.005583	1/min	SPR/Biacore (Biacore 3000)	A-Priori and Top Down	[1]
Golimumab	Monkey	kint	0.866	1/h	model estimated by the authors of the source	A-Priori and Top Down	[2]

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Golimumab	Monkey	Reference target concentration	0.001	nM	Model-derived TNF-alpha baseline assumption (1 pM)	A-Priori and Top Down	[3]
Golimumab	Monkey	kdeg	0.223	1/h	Model parameter from anti-TNF PBPK analysis	A-Priori and Top Down	[4]
Golimumab	Monkey	Kd (FcRn) in endosomal space	1.07	uM	Model-based meta-analysis	A-Priori	[5]
Nivolumab	Monkey	Kd	0.00145	uM	SPR assay (Biacore T100)	A-Priori and Top Down	[12]
Nivolumab	Monkey	koff	0.114	uM/1/min	SPR/Biacore	A-Priori and Top Down	[12]
Nivolumab	Monkey	kint	0.027	1/h	Model estimated by the authors	A-Priori and Top Down	[13]
Nivolumab	Monkey	Reference target concentration	0.0166	nM	Model estimated by the authors	A-Priori and Top Down	[13]

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Nivolumab	Monkey	kdeg	0.027	1/h	Model estimated by the authors	A-Priori and Top Down	[13]
Nivolumab	Monkey	Kd (FcRn) in endosomal space	1.07	uM	Model-based meta-analysis	A-Priori	[5]
Pembrolizumab	Monkey	Kd	0.00031	uM	SPR/Biacore	A-Priori and Top Down	[13]
Pembrolizumab	Monkey	koff	0.0017	1/min	SPR/Biacore	A-Priori and Top Down	[13]
Pembrolizumab	Monkey	kint	0.027		Model estimated by the authors	A-Priori and Top Down	[13]
Pembrolizumab	Monkey	Reference target concentration	0.0166	nM	Model estimated by the authors	A-Priori and Top Down	[13]
Pembrolizumab	Monkey	kdeg	0.027	1/h	Model estimated by the authors	A-Priori and Top Down	[13]
Pembrolizumab	Monkey	Kd (FcRn) in endosomal	0.081	uM	SPR/Biacore	A-Priori	[13]

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Tocilizumab	Monkey	Kd	0.00134	uM	SPR/Biacore	A-Priori and Top Down	[14]
Tocilizumab	Monkey	koff	0.01284	1/min	SPR/Biacore	A-Priori and Top Down	[14]
Tocilizumab	Monkey	kint	0.28	1/h	Assumed same as kdeg; Pulse chase experiments	A-Priori and Top Down	[15]
Tocilizumab	Monkey	Reference target concentration	2.2	nM	Ligand-binding receptor assay (assumed same across species)	A-Priori and Top Down	[16]
Tocilizumab	Monkey	kdeg	0.28	1/h	Pulse chase experiments	A-Priori and Top Down	[15]
Tocilizumab	Monkey	Kd (FcRn) in endosomal space	1.07	uM	Model-based meta analysis	A-Priori	[5]
Adalimumab	Human	Kd	0.000127	uM	SPR/Biacore (Biacore 3000)	A-Priori and Top Down	[1]

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Adalimumab	Human	koff	0.0096	1/min	SPR/Biacore (Biacore 3000)	A-Priori and Top Down[1]	
Adalimumab	Human	kint	0.866	1/h	Estimated from minimal PBPK model by the authors	A-Priori and Top Down[2]	
Adalimumab	Human	Reference target concentration	0.001	nM	Model fitting by the authors of [3]	A-Priori and Top Down[3]	
Adalimumab	Human	kdeg	0.223	1/h	Model parameter from anti-TNF PBPK analysis	A-Priori and Top Down[4]	
Adalimumab	Human	Kd (FcRn) in endosomal space	1.07	uM	Model-based meta-analysis	A-Priori[5]	
Atezolizumab	Human	Kd	0.299000299	uM	Equilibrium binding studies	A-Priori and Top Down[6]	
Atezolizumab	Human	koff	0.00936	1/min	SPR/Biacore	A-Priori and Top[6]	

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Atezolizumab	Human	kint	0.606	1/h	Flow cytometry internalization assay in BxPC-3 cells	A-Priori and Top Down	[6]
Atezolizumab	Human	Reference target concentration	0.609	nM	model fitting by the authors	A-Priori and Top Down	[6]
Atezolizumab	Human	kdeg	0.0426	1/h	Model estimated by the authors [6]	A-Priori and Top Down	[6]
Atezolizumab	Human	Kd (FcRn) in endosomal space	0.65	uM	SPR/Biacore	A-Priori	[6]
Avelumab	Human	Kd	0.0000467	uM	SPR assay (Biacore T200)	A-Priori and Top Down	[7]
Avelumab	Human	koff	0.00174	1/min	SPR assay (Biacore T200)	A-Priori and Top Down	[8]
Avelumab	Human	kint	0.25	1/h	pHrodo flow-cytometry internalization assay	A-Priori and Top Down	[7]

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Avelumab		Reference target concentration	0.609	nM	Model estimated by the authors [6]	A-Priori and Top Down [6]	[6]
	Human						
Avelumab		kdeg	0.0426	1/h	Model estimated by the authors	A-Priori and Top Down [6]	[6]
	Human						
Avelumab	Human	Kd (FcRn) in endosomal space	1.07	uM	Model-based meta-analysis	A-Priori [5]	
Bevacizumab	Human	Kd	0.000058	uM	SPR-Biacore	A-Priori and Top Down [9]	
Bevacizumab	Human	koff	0.00186	1/min	SPR-Biacore	A-Priori and Top Down [9]	
Bevacizumab	Human	kint	0.48	1/h	Assumed same as kdeg	A-Priori and Top Down [17]	
Bevacizumab	Human	Reference target concentration	0.013	nM	quantitative fluorescence (assumed same	A-Priori and Top [11]	

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Bevacizumab	Human	kdeg	0.48	1/h	cycloheximide-chase protein degradation assay; western blot dosimetry	A-Priori and Top Down	[17]
Bevacizumab	Human	Kd (FcRn) in endosomal space	1.07	uM	Model-based meta-analysis	A-Priori	[5]
Golimumab	Human	Kd	0.000018	uM	SPR/Biacore (Biacore 3000)	A-Priori and Top Down	[1]
Golimumab	Human	koff	0.005583	1/min	SPR/Biacore (Biacore 3000)	A-Priori and Top Down	[1]
Golimumab	Human	kint	0.866	1/h	Model estimated by the authors [2]	A-Priori and Top Down	[2]
Golimumab	Human	Reference target concentration	0.001	nM	Model-derived TNF-alpha baseline assumption (1 pM)	A-Priori and Top Down	[3]

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Golimumab	Human	kdeg	0.223	1/h	Model estimated by the authors [4]	A-Priori and Top Down	[4]
Golimumab	Human	Kd (FcRn) in endosomal space	1.07	uM	Model-based meta-analysis	A-Priori	[5]
Nivolumab	Human	Kd	0.00145	uM	SPR assay (BIAcore T100)	A-Priori and Top Down	[12]
Nivolumab	Human	koff	0.114	uM1/min	SPR/Biacore	A-Priori and Top Down	[12]
Nivolumab	Human	kint	0.027	1/h	Model estimated by the authors [13]	A-Priori and Top Down	[13]
Nivolumab	Human	Reference target concentration	0.07138	nM	Model estimated by the authors [13]	A-Priori and Top Down	[13]
Nivolumab	Human	kdeg	0.027	1/h	Model estimated by the authors [13]	A-Priori and Top Down	[13]
Nivolumab	Human	Kd (FcRn) in endosomal	1.07	uM	Model-based meta-analysis	A-Priori	[5]

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		space					
Pembrolizumab	Human	Kd	0.00031	uM	SPR/Biacore	A Priori and Top Down	[13]
Pembrolizumab	Human	koff	0.0017	1/min	SPR/Biacore	A Priori and Top Down	[13]
Pembrolizumab	Human	kint	0.027		Model estimated by the authors [13]	A Priori and Top Down	[13]
Pembrolizumab	Human	Reference target concentration	0.07138	nM	Model estimated by the authors [13]	A Priori and Top Down [13]	[13]
Pembrolizumab	Human	kdeg	0.027	1/h	Model estimated by the authors of the source	A Priori and Top Down [13]	[13]
Pembrolizumab	Human	Kd (FcRn) in endosomal space	0.081	uM	SPR/Biacore	A Priori [13]	[13]
Tocilizumab	Human	Kd	0.0013	uM	SPR/Biacore	A Priori	[14]

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b	n		4			and Top Down	
Tocilizuma b	Huma n	koff	0.0128 4	1/mi n	SPR/Biacore	A-Priori and Top Down	[14]
Tocilizuma b	Huma n	kint	0.28	1/h	Pulse chase experiments	A-Priori and Top Down	[15]
Tocilizuma b	Huma n	Reference target concentrati on	1	nM	Ligand-binding receptor assay	A-Priori and Top Down	[16]
Tocilizuma b	Huma n	kdeg	0.28	1/h	Pulse chase experiments	A-Priori and Top Down	[15]
Tocilizuma b	Huma n	Kd (FcRn) in endosomal space	1.07	uM	Model-based meta- analysis	A-Priori	[5]

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S1.4. Top-down Linear Clearance Estimates for Monkey

Molecule	CL (L/h/kg) [RSE %]	V1 (L/kg) [RSE %]	Q (L/h/kg) [RSE %]	V2 (L/kg) [RSE %]
Adalimumab	0.0004 [14.9]	0.0467 [11.1]	0.0010 [Fixed]	0.0276 [Fixed]
Atezolizumab	0.0002 [8.1]	0.0296 [8.5]	0.0010 [Fixed]	0.0221 [23.4]
Avelumab	0.0012 [1.8]	0.0435 [6.2]	0.0015 [21.8]	0.0394 [8.6]
Bevacizumab	0.0003 [14.5]	0.0474 [5.6]	0.0010 [Fixed]	0.0270 [31.2]
Gotimumab	0.0004 [5.9]	0.0288 [3.2]	0.0038 [24.4]	0.0171 [9.8]
Nivolumab	0.0002 [5.4]	0.0312 [8.5]	0.0010 [Fixed]	0.0149 [32.8]
Pembrolizumab	0.0002 [18.6]	0.0150 [23.8]	0.0007 [23.5]	0.0254 [22.5]
Tocilizumab	0.0008 [5.2]	0.0477 [3.8]	0.0010 [Fixed]	0.0206 [28.6]

S1.5. Percent Extrapolated in AUCinf Calculations in Observed Data:

Species	Molecule	Dose (mg/kg)	% Extrapolated in AUC _{inf}
Monkey	Adalimumab	3	29
		0.5	37
		5	13
		20	31
	Avelumab	5	5
		4	22
	Bevacizumab	5	5
		3	16
	Nivolumab	1	21
		10	20
	Pembrolizumab	0.3	28
		3	59
		30	38
Human	Adalimumab	10	15
		0.5	25
		1	30
		3	30
		5	31
		10	32

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	Atezolizumab	1	31
		3	25
		10	31
		15	31
		20	35
	Avelumab	1	11
		3	9
		10	24
		20	15
	Bevacizumab	15	33
	Golimumab	0.1	14
		0.3	15
		1	19
		3	19
		6	23
		10	31
	Nivolumab	1	50
		3	45
		4	45
		6	48
		10	42
		20	70
	Pembrolizumab	2	42
		10	35
	Tocilizumab	8	0

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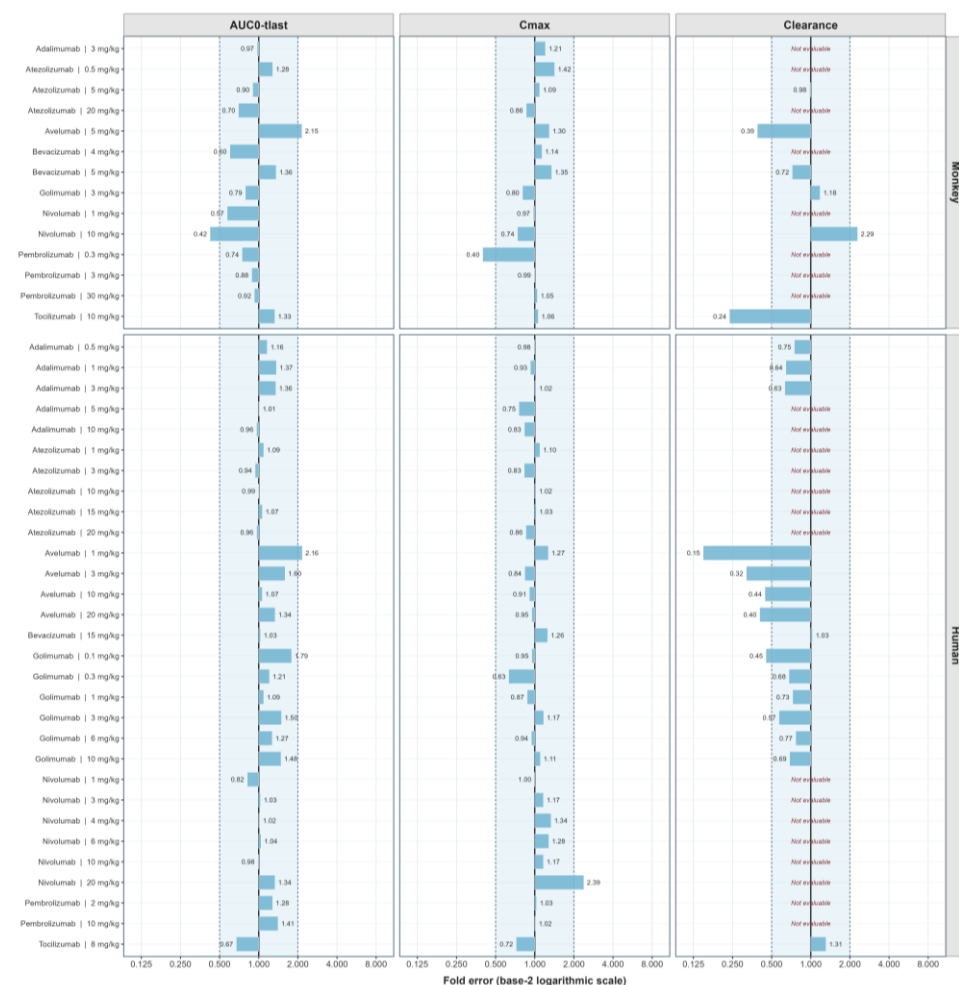
S1.6

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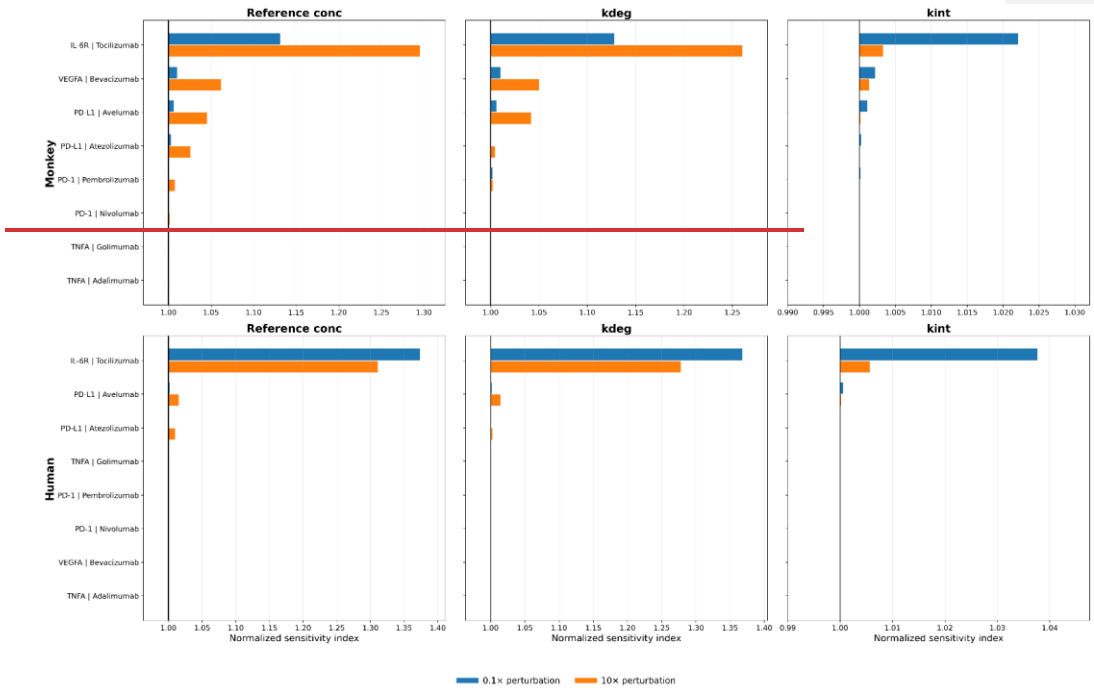
S2.2. Fold errors from PBPK model predictions without calibration with in vivo data. Directional fold errors are shown for $AUC_{0-t_{last}}$, C_{max} , and clearance across the evaluated molecule-dose scenarios. Fold error was calculated as the predicted value divided by the observed value. Horizontal bars extend from the line of unity on a base-2 logarithmic scale; values below and above 1 indicate underprediction and overprediction, respectively. The shaded region and dashed lines denote the prespecified 0.5- to 2-fold acceptance interval. Clearance scenarios were marked not evaluable when no observed profile had an AUC_{inf} extrapolated fraction of 20% or less. Models were not calibrated using in vivo pharmacokinetic data.

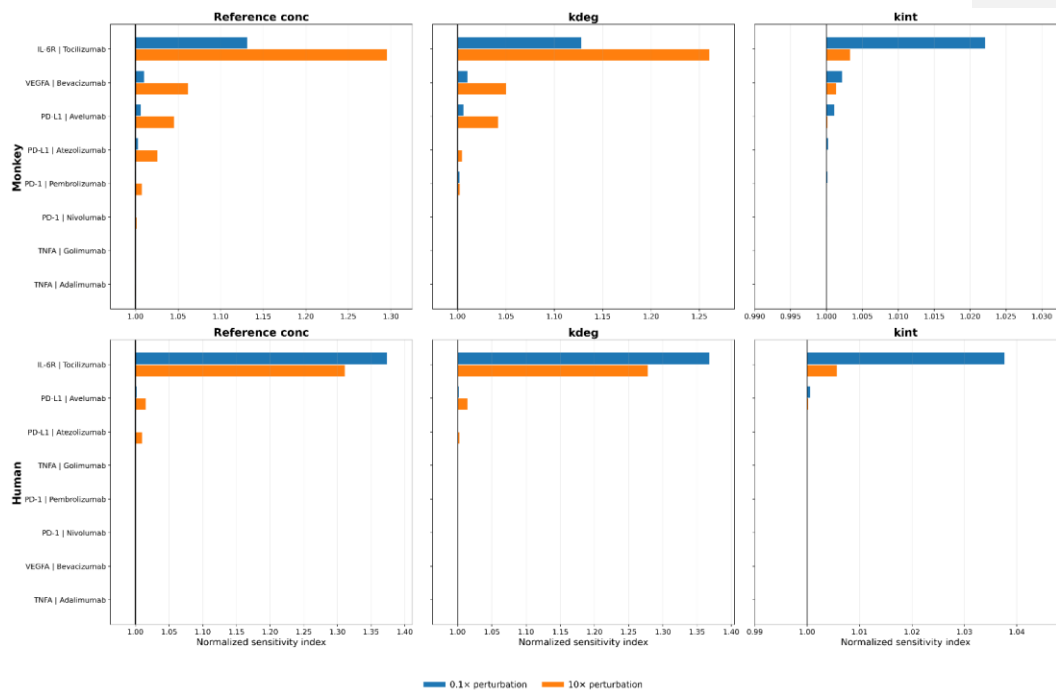


S2.3. Local Sensitivity Analysis of Target Related Parameters on Clearance Estimate

Using PBPK Model. An a priori PBPK modeling framework was used to assess parameter sensitivity by varying the target reference concentration, target degradation rate (kdeg), and internalization rate (kint) one at a time while holding all other model parameters constant. Each parameter was varied by decreasing to 0.1-fold and increasing to 10-fold of its reference value, and the resulting terminal clearance was compared with the reference simulation. The normalized sensitivity coefficient was calculated as $\ln(CL_{varied}/CL_{reference})/\ln(P_{varied}/P_{reference})$, and a sensitivity index of 1+ the

normalized coefficient was reported, such that a value of 1 indicates no sensitivity. Sensitivity indices were averaged across evaluated doses for each molecule and ranked from highest to lowest in the figure.





References

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1. Shealy DJ, Cai A, Staquet K, Baker A, Lacy ER, Johns L, et al. Characterization of golimumab, a human monoclonal antibody specific for human tumor necrosis factor alpha. *mAbs*. 2010;2(4):428-439. doi:10.4161/mabs.2.4.12304.
2. Zheng S, Niu J, Geist B, Fink D, Xu Z, Zhou H, Wang W. A minimal physiologically based pharmacokinetic model to characterize colon TNF suppression and treatment effects of an anti-TNF monoclonal antibody in a mouse inflammatory bowel disease model. *mAbs*. 2020;12(1):1813962. doi:10.1080/19420862.2020.1813962.
3. Reig-López J, García-Otero X, Cuquerella-Gilabert M, Bandín-Vilar E, Otero-Espinar FJ, Fernández-Ferreiro A, et al. A physiologically based pharmacokinetic model to describe [89Zr]Zr-DFO-adalimumab specific activity after intravitreal administration to rats with endotoxin-induced uveitis. *International Journal of Pharmaceutics*. 2025;681:125845. doi:10.1016/j.ijpharm.2025.125845.
4. Chen J, Lin R, Guo G, Wu W, Ke M, Ke C, et al. Physiologically-Based Pharmacokinetic Modeling of Anti-Tumor Necrosis Factor Agents for Inflammatory Bowel Disease Patients to Predict the Withdrawal Time in Pregnancy and Vaccine Time in Infants. *Clinical Pharmacology & Therapeutics*. 2023;114(6):1254-1263. doi:10.1002/cpt.3031.
5. Benamara S, Sjögren E, Gattacceca F, Chenel M, Deslandes A, Nguyen L, Teutonico D. Prediction of Monoclonal Antibodies Pharmacokinetics in Human: Identification of a Reference Neonatal Fc Receptor (FcRn) Binding Affinity Using Physiologically Based Pharmacokinetic (PBPK) Modeling. *ACS Pharmacology & Translational Science*. 2026;9:177-190. doi:10.1021/acsptsci.5c00674.
6. Huang W, Stader F, Chan P, Shemesh CS, Chen Y, Gill KL, et al. Development of a pediatric physiologically-based pharmacokinetic model to support recommended dosing of atezolizumab in children with solid tumors. *Frontiers in Pharmacology*. 2022;13:974423. doi:10.3389/fphar.2022.974423.
7. Jin H, D'Urso V, Neuteboom B, McKenna SD, Schweickhardt R, Gross AW, et al. Avelumab internalization by human circulating immune cells is mediated by both Fc gamma receptor and PD-L1 binding. *Onc Immunology*. 2021;10(1):1958590. doi:10.1080/2162402X.2021.1958590.
8. Benicky J, Sanda M, Brnakova Kennedy Z, Grant OC, Woods RJ, Zwart A, Goldman R. PD-L1 Glycosylation and Its Impact on Binding to Clinical Antibodies. *Journal of Proteome Research*. 2021;20(1):485-497. doi:10.1021/acs.jproteome.0c00521.

9. Papadopoulos N, Martin J, Ruan Q, Rafique A, Rosconi MP, Shi E, et al. Binding and neutralization of vascular endothelial growth factor (VEGF) and related ligands by VEGF Trap, ranibizumab and bevacizumab. *Angiogenesis*. 2012;15(2):171-185. doi:10.1007/s10456-011-9249-6.
10. Calera MR, Venkatakrishnan A, Kazlauskas A. VE-cadherin increases the half-life of VEGF receptor 2. *Experimental Cell Research*. 2004;300(1):248-256. doi:10.1016/j.yexcr.2004.07.007.
11. Imoukhuede PI, Popel AS. Quantification and cell-to-cell variation of vascular endothelial growth factor receptors. *Experimental Cell Research*. 2011;317(7):955-965. doi:10.1016/j.yexcr.2010.12.014.
12. Tan S, Zhang H, Chai Y, Song H, Tong Z, Wang Q, et al. An unexpected N-terminal loop in PD-1 dominates binding by nivolumab. *Nature Communications*. 2017;8:14369. doi:10.1038/ncomms14369.
13. Lindauer A, Valiathan CR, Mehta K, Sriram V, de Greef R, Elassaiss-Schaap J, de Alwis DP. Translational Pharmacokinetic/Pharmacodynamic Modeling of Tumor Growth Inhibition Supports Dose-Range Selection of the Anti-PD-1 Antibody Pembrolizumab. *CPT Pharmacometrics & Systems Pharmacology*. 2017;6(1):11-20. doi:10.1002/psp4.12130.
14. Xu C, Rafique A, Potocky T, Paccaly A, Nolain P, Lu Q, et al. Differential Binding of Sarilumab and Tocilizumab to IL-6R α and Effects of Receptor Occupancy on Clinical Parameters. *Journal of Clinical Pharmacology*. 2021;61(5):714-724. doi:10.1002/jcph.1795.
15. Gerhartz C, Dittrich E, Stoyan T, Rose-John S, Yasukawa K, Heinrich PC, Graeve L. Biosynthesis and half-life of the interleukin-6 receptor and its signal transducer gp130. *European Journal of Biochemistry*. 1994;223(1):265-274. doi:10.1111/j.1432-1033.1994.tb18991.x.
16. Nesbitt JE, Fuller GM. Differential regulation of interleukin-6 receptor and gp130 gene expression in rat hepatocytes. *Molecular Biology of the Cell*. 1992;3(1):103-112. doi:10.1091/mbc.3.1.103.
17. Sarabipour S, Kinghorn K, Quigley KM, Kovacs-Kasa A, Annex BH, Bautch VL, Mac Gabhann F. Trafficking dynamics of VEGFR1, VEGFR2, and NRP1 in human endothelial cells. *PLOS Computational Biology*. 2024;20(2):e1011798. doi:10.1371/journal.pcbi.1011798.

Supplementary Material 2. Github link to full project codes: <https://github.com/namlink-org/APriori-PBPK-mAb>